



Indian Council of Medical Research

ICMR AMRIT

Operator manual

One system, two programs. A laboratory runs the desktop application; a country runs the central server. Between them travels nothing but aggregate numbers. Every screenshot in this manual is the real software, captured from a running installation.

Deployment

India

Contract version

2.0

Issued

2026-08-14

Antimicrobial Resistance Monitoring, Intelligence and Tracking

This manual covers both halves of the system and the reasoning behind them:

- **Part I — the desktop application**, which a laboratory runs on its own computer, offline.
- **Part II — the central web portal**, which a programme runs for a country or a network.
- **Part III — methods and rationale**, which states what each analytical choice is, what the alternatives were, and why this one was taken. Part III is written to be quotable in a paper: every method names the module that implements it.

Every screenshot is captured from a running build — the desktop screens by the application's own `AMRIT_CAPTURE_DIR` mode against a freshly seeded demonstration database, the portal screens by `tools/capture_portal.mjs` signed in as each role — and every figure is generated by `tools/build_manual_figures.py`. Neither is drawn by hand, because a picture that is edited separately from the system drifts away from it and nobody notices until a reader follows it and is wrong.

Part II — Central web portal 13. Deploy the server 14. The site registry and enrolment 15. Asking questions 16. Dashboards, by stakeholder 17. The outbreak console 18. From a signal to an action plan 19. The public view 20. Audit and governance 21. Portal administration: users, roles, capabilities

Part III — Methods and rationale 22. Data model and standards 23. Breakpoint interpretation 24. Deduplication 25. Indicators 26. Outbreak detection 27. Privacy engineering 28. Federation protocol design 29. Dashboard computation 30. Limitations and threats to validity

Appendices — A commands · B environment · C role capabilities · D metric catalogue · E demo credentials

Part I — Desktop application

1. What it is for, and what it refuses to do

A microbiology laboratory produces isolate records: a patient, a specimen, an organism, and a panel of antimicrobial susceptibility results. AMRIT's desktop application is where those records live. It is a local-first application — Electron over SQLite — and it is fully functional with no network at all. A laboratory can install it, type records for a year, analyse them, print an antibiogram and export a WHONET file without ever connecting to anything.

It refuses three things by construction:

- It never uploads a patient row. What leaves this application over the network is a count or a rate (§27, §28).
- It never interprets a susceptibility result against a breakpoint set nobody activated (§23).

- It never invents data. A value it cannot read, map or interpret is reported and left blank, not guessed.

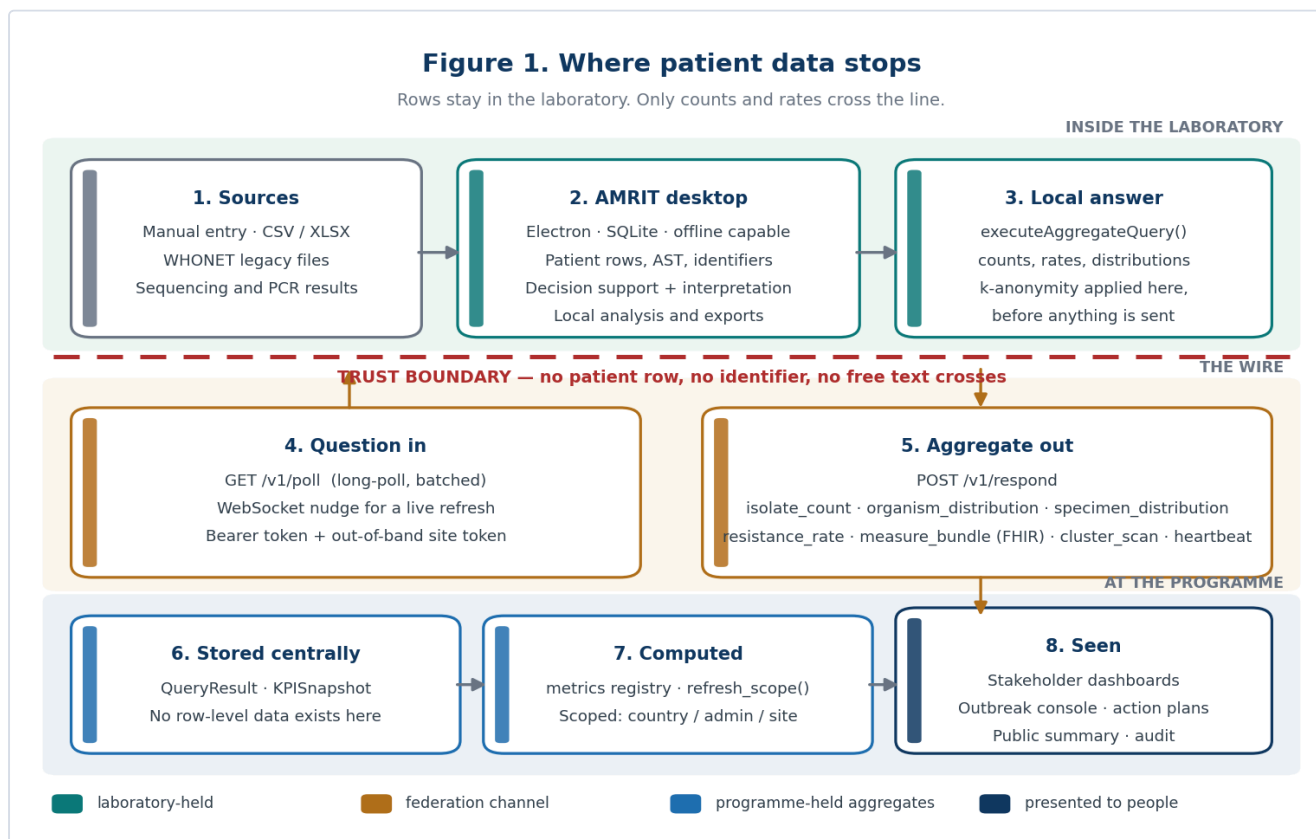
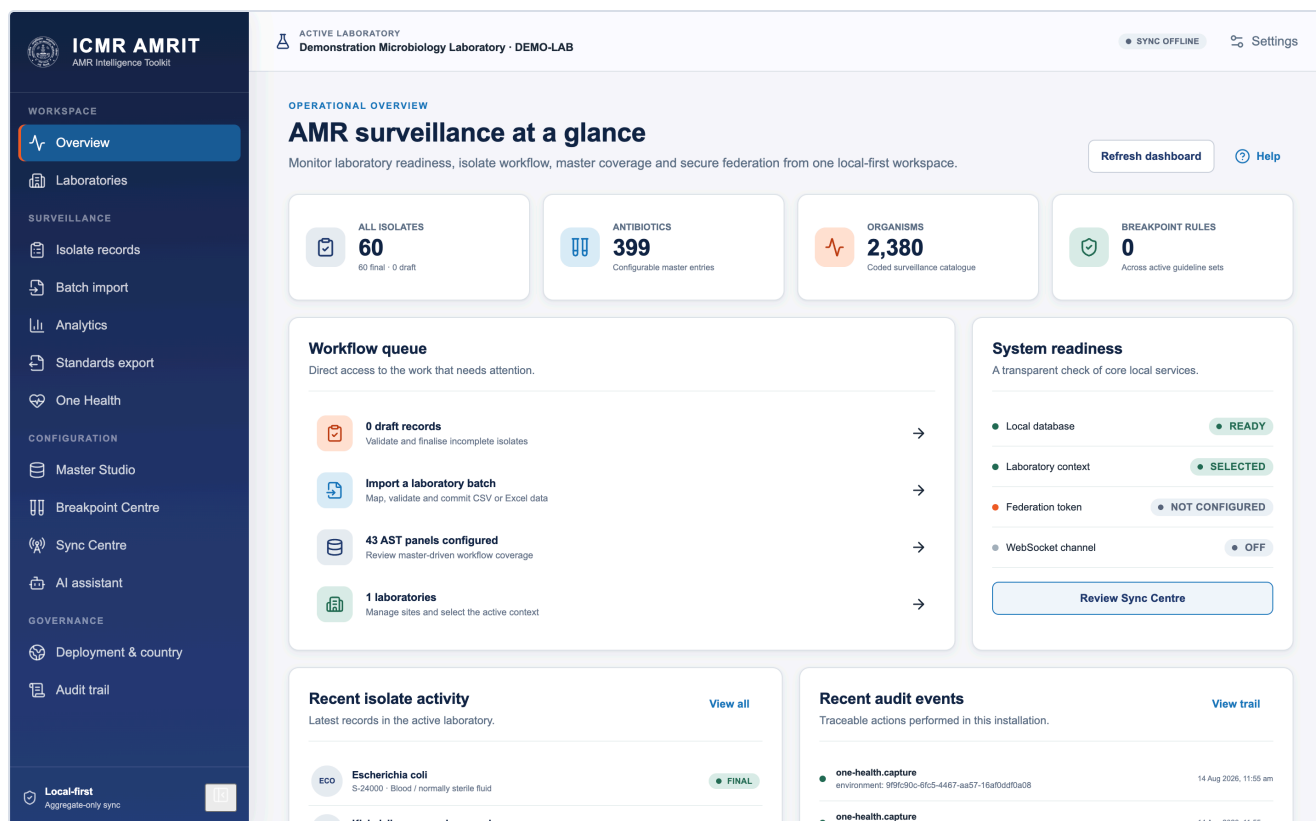


Figure 1 — where patient data stops.



2. Install and first run

Install the build for your platform (`.dmg` , `.exe` , `.AppImage`). On first run the application creates its database under the operating system's application-data directory and seeds the packaged catalogue: WHONET organisms, antimicrobials, specimen types, expert rules and panels, hash-pinned so a tampered catalogue is refused rather than loaded.

Nothing else is created. There is no default account, no default laboratory and no sample data — a deployment holding real results must never inherit invented ones.

3. Configure the deployment

Deployment → country profile. The profile decides the country, its administrative tiers (what they are called and how many there are), the default guideline (CLSI or EUCAST), identifier namespaces, branding and the retention period. A profile is data, not code: the same binary serves any country, and `python3 tools/generate_country_reference.mjs` scaffolds a new one.

The screenshot shows the 'Deployment and country' configuration page in the ICMR AMRIT application. The left sidebar contains a navigation menu with sections: WORKSPACE (Overview, Laboratories), SURVEILLANCE (Isolate records, Batch import, Analytics, Standards export, One Health), CONFIGURATION (Master Studio, Breakpoint Centre, Sync Centre, AI assistant), and GOVERNANCE (Deployment & country, Audit trail). The 'Deployment & country' section is highlighted. The main content area is titled 'Deployment and country' and includes a subtitle: 'Country-varying behaviour for India (IND). Changes take effect immediately; no restart and no file editing.' There are three buttons: 'Import', 'Export', and 'Save changes'. The 'Base country' section has a dropdown for 'Deployment country' set to 'India'. The 'Identity and branding' section includes fields for 'Country name' (India), 'Product name' (ICMR AMRIT), and 'Authority name' (Indian Council of Medical Research). There is a 'Logo' section with a 'Choose a logo...' button and a 'Brand' section with three color swatches: Brand navy, Brand blue, and Brand orange.

Two changes are irreversible and say so before they are saved: changing the identifier namespace (existing identifiers keep the old one) and changing the country (geography and overrides from the previous country are discarded).

4. Register the laboratory and place it on the map

Laboratories → New laboratory. A laboratory needs a code, a name and an address in the country's own address shape — the form is generated from the profile, so an Indian deployment asks for state and district and a Testland deployment asks for governorate and district.

Add laboratory

Required identity and default AST settings. Location codes can also be managed in Master Studio.

Laboratory code *

Laboratory name *

Stable identifier used in records and federation.

Country *

India

The ISO 3166-1 alpha-3 code is stored with the name.

ISO country code

IND

ISO 3166-1 alpha-3, filled from the country and stored with every record.

POSTAL ADDRESS

The address as it is written on an envelope in this country. Which fields appear, what they are called and the order they are asked in all come from the country, not from this form.

Plus Code

For example 7J3Q2M8Q+P9

Optional full Google Open Location Code. It works offline and can place a facility where postal or street addressing is unavailable. Short local codes are not accepted.

Organisation

Street address *

City *

PIN code *

For example 110034, 110001

No coordinates yet

Cancel Save laboratory

The address resolves to an administrative unit from the bundled geo pack. That unit — not a typed place name — is what every scope filter, dashboard roll-up and map placement uses afterwards.

Edit laboratory

Required identity and default AST settings. Location codes can also be managed in Master Studio.

Laboratory code *

DEMO-LAB

Stable identifier used in records and federation.

Laboratory name *

Demonstration Microbiology Laboratory

Country *

India

The ISO 3166-1 alpha-3 code is stored with the name.

ISO country code

IND

ISO 3166-1 alpha-3, filled from the country and stored with every record.

POSTAL ADDRESS

The address as it is written on an envelope in this country. Which fields appear, what they are called and the order they are asked in all come from the country, not from this form.

Plus Code

For example 7J3Q2M8Q+P9

Optional full Google Open Location Code. It works offline and can place a facility where postal or street addressing is unavailable. Short local codes are not accepted.

Organisation

Street address *

City *

PIN code *

For example 110034, 110001

No coordinates yet

Cancel Save laboratory

Site coordinates are optional and consent-gated (Sync → Optional location sharing). With consent given, **Use this computer's location** reads the operating system's location service; the coordinates can always be typed instead, and the source (device or hand-entered) travels with the heartbeat, because how a coordinate was obtained is part of what it means.

5. Catalogues and breakpoints

Master Studio holds the reference data: organisms, antimicrobials, specimens, panels, expected-resistance rules, expert rules, and local additions. Packaged rows are read-only; local rows are marked custom and can be deactivated but never silently overwrite the catalogue.

The screenshot displays the ICMR AMRIT Master Studio interface. The left sidebar contains navigation menus for Workspace, Surveillance, Configuration, and Governance. The main panel shows the 'Master Studio' configuration page for 'Antibiotics'. A table lists various antibiotics with their codes, names, classes, subclasses, WHO awareness, status, and actions.

CODE	NAME	CLASS	SUBCLASS	WHO AWARE	STATUS	ACTIONS
AMC	Amoxicillin/Clavulanic acid	Beta-lactam+Inhibitors		Access	ACTIVE	Edit Duplicate Disable
APX	Aspoxicillin	Penicillins		Watch	ACTIVE	Edit Duplicate Disable
BDP	Brodinoprim	Folate pathway inhibitors		Access	ACTIVE	Edit Duplicate Disable
BUT	Butoconazole	Antifungals			ACTIVE	Edit Duplicate Disable
CAP	Capreomycin	Antimycobacterials			ACTIVE	Edit Duplicate Disable
CRB	Carbenicillin	Penicillins	Carboxypenicillin	Watch	ACTIVE	Edit Duplicate Disable
CAR	Carumonam	Monobactams		Reserve	ACTIVE	Edit Duplicate Disable
CAS	Caspofungin	Antifungals			ACTIVE	Edit Duplicate Disable
CAC	Cefacetrile	Cephems	Cephalosporin I	Access	ACTIVE	Edit Duplicate Disable

Breakpoint Centre is where guidelines arrive. EUCAST publishes its tables free of charge and permits redistribution, so the application fetches and stages the current edition with one button. CLSI's M100 is a paid standard: the application links to it and imports a workbook you supply. Either way the set is *staged inactive* and must be activated deliberately (\$23).

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AMR Intelligence Toolkit

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Demonstration Microbiology Laboratory · DEMO-LAB

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INTERPRETATION GOVERNANCE

Breakpoint Centre

Acquire, stage, inspect and activate versioned breakpoint sources while preserving edition and file provenance.

+ Custom row

Upload workbook

Download official toolkit

Help

Clinical governance boundary

Imported workbooks are staged as versioned sets. Local custom rows are saved inactive and cannot affect interpretation until reviewed and activated through the governed workflow. Re-reading a source replaces its own earlier staging only while that staging has never been activated. Existing result provenance is never rewritten.

EUCAST

EUCAST clinical breakpoints

Published free of charge and redistributable with attribution, so this is the one table AMRIT can fetch for you. The download is staged for review, never activated automatically.
Installed: v15.0, 950 rows. Run the update when EUCAST publishes a new edition.

Update EUCAST breakpoints

Open eucast.org

CLSI

Breakpoint Implementation Toolkit

Open the current official implementation resources, then download the supported public workbook through the secure service.

Open official toolkit

Access M100

Structured local workbook

Upload .xlsx, .xlsb or .xls. A normalized "Breakpoints" sheet can include guideline, edition, method, organism, antibiotic, R/I/S thresholds, unit and comments.

Choose workbook

Versioned breakpoint sets

0 active sets. Activation is an explicit, audited operation.

SOURCE SET	STATUS	ROWS	VALIDATION	PROVENANCE	IMPORTED	ACTIONS
EUCAST Breakpoint Tables v15.0 15.0	STAGED	950	READY 0 unmatched codes	c54008889008df1e...	14 Aug 2026, 6:26 am	Activate

Breakpoint master preview

Showing 100 of 950 customizable interpretation rows.

The screenshot above is a live fetch: EUCAST v15.0, 950 rows staged, **0 unmatched codes**, **READY**, and an **Activate** button that has not been pressed. Until it is, nothing in this set interprets anything.

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GUIDELINE	EDITION	METHOD	ORGANISM	ANTIBIOTIC	SITE · ROUTE	R	I	S
EUCAST	15.0	MIC	Enterobacterales	AMP	iv	8	—	8
EUCAST	15.0	Disk diffusion	Enterobacterales	AMP	iv	14	—	14
EUCAST	15.0	MIC	Enterobacterales	AMP	Uncomplicated urinary tract infection · oral	8	—	8
EUCAST	15.0	Disk diffusion	Enterobacterales	AMP	Uncomplicated urinary tract infection · oral	14	—	14
EUCAST	15.0	MIC	Enterobacterales	SAM	iv	8	—	8
EUCAST	15.0	Disk diffusion	Enterobacterales	SAM	iv	14	—	14
EUCAST	15.0	MIC	Enterobacterales	SAM	Uncomplicated urinary tract infection · oral	8	—	8
EUCAST	15.0	Disk diffusion	Enterobacterales	SAM	Uncomplicated urinary tract infection · oral	14	—	14
EUCAST	15.0	MIC	Enterobacterales	AMX	iv	8	—	8
EUCAST	15.0	MIC	Enterobacterales	AMX	Urinary tract · oral	8	—	0.001
EUCAST	15.0	MIC	Enterobacterales	AMX	Uncomplicated urinary tract infection · oral	8	—	8
EUCAST	15.0	MIC	Enterobacterales	AMX	oral	8	—	0.001
EUCAST	15.0	MIC	Enterobacterales	AMC	iv	8	—	8
EUCAST	15.0	Disk diffusion	Enterobacterales	AMC	iv	19	—	19
EUCAST	15.0	MIC	Enterobacterales	AMC	Urinary tract · oral	8	—	0.001
EUCAST	15.0	Disk diffusion	Enterobacterales	AMC	Urinary tract · oral	19	—	50
EUCAST	15.0	MIC	Enterobacterales	AMC	Uncomplicated urinary tract infection · oral	32	—	32
EUCAST	15.0	Disk diffusion	Enterobacterales	AMC	Uncomplicated urinary tract infection · oral	16	—	16
EUCAST	15.0	MIC	Enterobacterales	AMC	oral	8	—	0.001
EUCAST	15.0	Disk diffusion	Enterobacterales	AMC	oral	19	—	50

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The preview lists what was staged — organism scope, agent, method, site and route, and the R/I/S thresholds — so a reviewer checks rows rather than trusting a row count.

6. Enter data

Three routes, all landing in the same validated shape:

Type a record. Identity first (patient identifier, specimen number and date, specimen type, organism), then residence, then AST. The antimicrobial panel offered is the one matching the organism and specimen, from the panel catalogue.

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SURVEILLANCE WORKFLOW

Isolate records

Enter, validate and review antimicrobial susceptibility records with master-driven panels and duplicate safeguards.

+ New Isolate Help

60 FINAL 0 DRAFT 60 SHOWN

Q Search specimen, patient or organism... All statuses Refresh

SPECIMEN	PATIENT ID	ORGANISM	LOCATION	AST	STATUS	ACTIONS
S-24000 14 Aug 2026 - Blood / normally sterile fluid	D-1000	Escherichia coli ECO	ICU ICU	7 results	FINAL	Review Export
S-24001 13 Aug 2026 - Urine	D-1001	Klebsiella pneumoniae complex KPN	Medicine Inpatient	6 results	FINAL	Review Export
S-24002 12 Aug 2026 - Lower respiratory tract	D-1002	Staphylococcus aureus SAU	Surgery Inpatient	8 results	FINAL	Review Export
S-24003 11 Aug 2026 - Deep tissue / wound / musculoskeletal	D-1003	Escherichia coli ECO	Paediatrics Inpatient	7 results	FINAL	Review Export
S-24004 10 Aug 2026 - Blood / normally sterile fluid	D-1004	Klebsiella pneumoniae complex KPN	Emergency ICU	6 results	FINAL	Review Export
S-24005 9 Aug 2026 - Urine	D-1005	Staphylococcus aureus SAU	ICU Inpatient	8 results	FINAL	Review Export
S-24006 8 Aug 2026 - Lower respiratory tract	D-1006	Escherichia coli ECO	Medicine Inpatient	7 results	FINAL	Review Export
S-24007 7 Aug 2026 - Deep tissue / wound / musculoskeletal	D-1007	Klebsiella pneumoniae complex KPN	Surgery Inpatient	6 results	FINAL	Review Export
S-24008 6 Aug 2026 - Blood / normally sterile fluid	D-1008	Staphylococcus aureus SAU	Paediatrics ICU	8 results	FINAL	Review Export
S-24009 5 Aug 2026 - Urine	D-1009	Escherichia coli ECO	Emergency Inpatient	7 results	FINAL	Review Export

The editor is where identity, specimen, organism and the antimicrobial panel are entered, and where decision support answers as you type:

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EDITOR.RECORDNUMBER

editor.reviewTitle

editor.purpose

Help

Specimen identity

Minimum coded identity used for duplicate detection and downstream exports.

Patient identifier *

D-1000

Local identifier; avoid direct names.

Specimen number *

S-24000

Specimen date *

14/08/2026

Specimen type

Blood / normally sterile fluid

Organism

Escherichia coli

Location type

ICU

One Health domain

Select...

Hospital / facility

Select...

Location / ward

ICU

Sex

f - Female

Date of birth

dd/mm/yyyy

Age in years

5

Admission date

dd/mm/yyyy

Diagnosis (uncoded detail)

Only for a diagnosis with no code in the catalogue. Coded diagnoses are what analyses group by.

Diagnosis

Search by code or by wording. Several diagnoses can be recorded; each keeps its code system.

No diagnosis coded.

Search diagnoses by code or wording...

Add

Other bacterial intestinal infections

A04 · ICD-10

Import a batch. Map columns once, save the mapping as a profile, and reuse it. The preview shows what will be imported, what will land as a draft and what is refused, per row and with a reason, before anything is written.

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DATA ONBOARDING

Batch import

Map and validate CSV or Excel laboratory data before an atomic commit to the active site.

Download template Choose file Help

Destination: Demonstration Microbiology Laboratory

All accepted rows will be assigned to DEMO-LAB. A failed commit rolls back the entire batch; the source file is never modified.



Select a CSV or Excel workbook

Supported: .xlsx, .xls, .xlsb, .csv, .tsv and delimited text. Protected or malformed workbooks are reported without partial ingestion.

Browse files

or start with the master-aware template

Import history

Recent auditable import operations for this installation.



No import history

Completed and failed import attempts will be recorded here.

WHONET legacy. Existing WHONET data files import directly, including their code sets.

7. Quality control and decision support

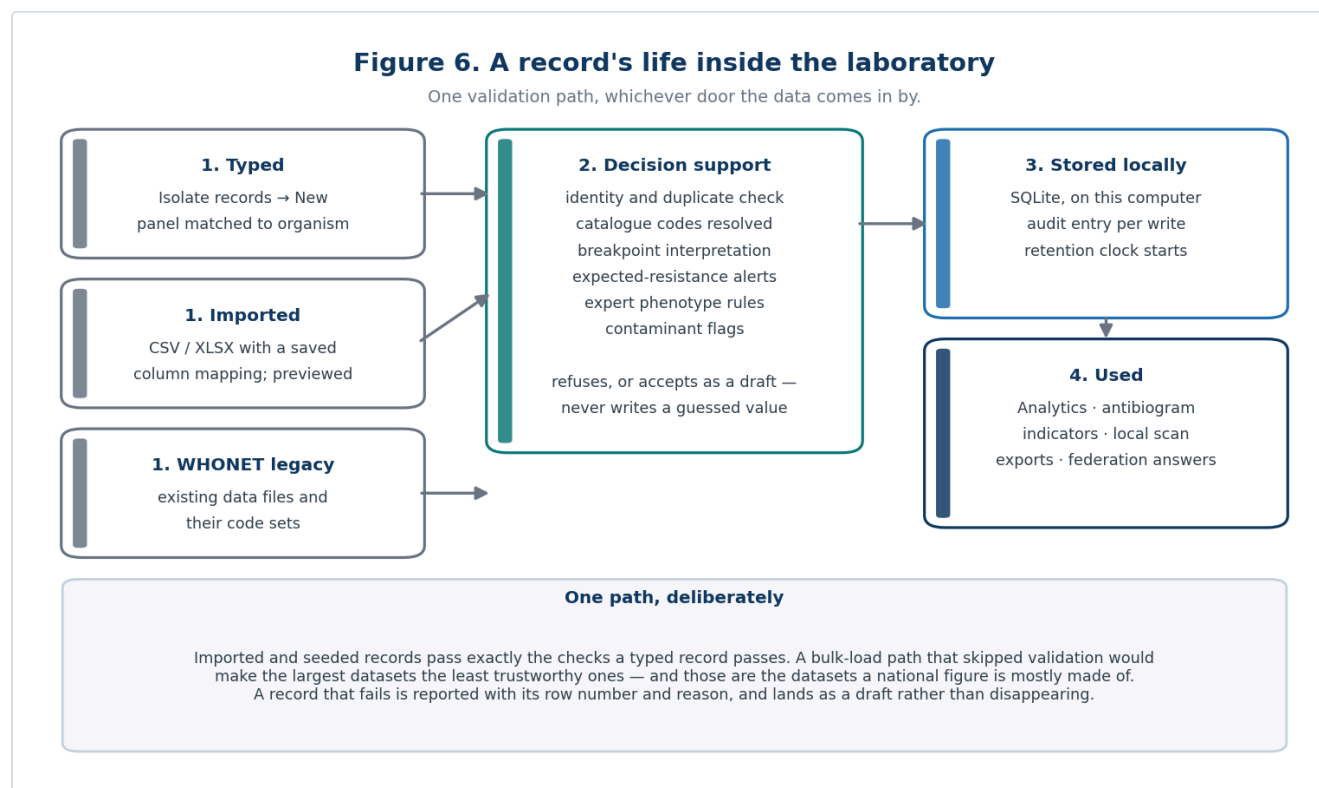


Figure 6 — one validation path, whichever door the data comes in by.

Every write passes the same decision-support layer, whether typed, imported or seeded:

- **Interpretation.** A measurement with no explicit S/I/R is interpreted from the active breakpoint set — and left blank with a stated reason when the set has no matching row, when two equally specific rows disagree, or when the record does not state something the breakpoint depends on (§23).
- **Expected resistance.** An organism reported susceptible to an agent it is intrinsically resistant to raises an alert rather than being corrected silently.
- **Expert rules.** Configurable phenotype rules (ESBL, carbapenemase, MRSA, inducible clindamycin) add comments and alerts.
- **Contaminant and duplicate checks.** Likely skin flora in a sterile-site culture is flagged; an identical patient/specimen identity is refused as a duplicate.

8. Analyse

Analytics runs deterministic, reproducible analyses over the local database: resistance rates by organism and agent, antibiograms, specimen and organism distributions, priority indicators, time trends, and a local outbreak scan.

The screenshot displays the ICMR AMRIT AMR analytics interface. The left sidebar contains navigation menus for WORKSPACE, SURVEILLANCE, CONFIGURATION, and GOVERNANCE. The main panel is titled 'AMR analytics' and includes a 'Run analysis' button. Below the title, the 'Analysis cohort' section shows filters applied to the active laboratory. The 'Analysis mode' is set to 'R / I / S distribution'. The 'Saved analysis macro' is 'No saved macro'. The 'Period start' is '14/08/2025' and the 'Period end' is '14/08/2026'. The 'Organism' is 'All organisms' and the 'Specimen' is 'All specimens'. The 'Antibiotic' is 'All antibiotics' and the 'Location type' is 'All locations'. Below these filters, four summary cards are displayed: 'ISOLATES IN COHORT' (60), 'RESISTANT RESULTS' (183), 'INTERMEDIATE' (46), and 'RESISTANCE' (43.6%). The bottom section contains 'Resistance interpretation' and 'Monthly volume' cards.

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ANALYSIS WORKSPACE

AMR analytics

Explore descriptive resistance patterns from validated local records with transparent filters and denominators.

[Run analysis](#) [Help](#)

Analysis cohort

Filters are applied to the active laboratory before denominators and percentages are calculated.

Analysis mode: R / I / S distribution
Core susceptibility interpretation summary.

Saved analysis macro: No saved macro
[Save current filters](#) [Delete macro](#)

Period start: 14/08/2025
Period end: 14/08/2026
Organism: All organisms
Specimen: All specimens
Antibiotic: All antibiotics
Location type: All locations

ISOLATES IN COHORT: 60
RESISTANT RESULTS: 183
INTERMEDIATE: 46
RESISTANCE: 43.6%

Resistance interpretation

Distribution across all AST interpretations in the filtered cohort.

Monthly volume

Included isolate records by specimen month.

With an analysis run, the same screen carries the antibiogram, the priority indicators and the resistance table computed over the current filter set:

This screenshot is identical to the one above, showing the ICMR AMRIT AMR analytics interface. The left sidebar contains navigation menus for WORKSPACE, SURVEILLANCE, CONFIGURATION, and GOVERNANCE. The main panel is titled 'AMR analytics' and includes a 'Run analysis' button. Below the title, the 'Analysis cohort' section shows filters applied to the active laboratory. The 'Analysis mode' is set to 'R / I / S distribution'. The 'Saved analysis macro' is 'No saved macro'. The 'Period start' is '14/08/2025' and the 'Period end' is '14/08/2026'. The 'Organism' is 'All organisms' and the 'Specimen' is 'All specimens'. The 'Antibiotic' is 'All antibiotics' and the 'Location type' is 'All locations'. Below these filters, four summary cards are displayed: 'ISOLATES IN COHORT' (60), 'RESISTANT RESULTS' (183), 'INTERMEDIATE' (46), and 'RESISTANCE' (43.6%). The bottom section contains 'Resistance interpretation' and 'Monthly volume' cards.

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Analysis cohort

Filters are applied to the active laboratory before denominators and percentages are calculated.

Analysis mode: R / I / S distribution
Core susceptibility interpretation summary.

Saved analysis macro: No saved macro
[Save current filters](#) [Delete macro](#)

Period start: 14/08/2025
Period end: 14/08/2026
Organism: All organisms
Specimen: All specimens
Antibiotic: All antibiotics
Location type: All locations

ISOLATES IN COHORT: 60
RESISTANT RESULTS: 183
INTERMEDIATE: 46
RESISTANCE: 43.6%

Resistance interpretation

Distribution across all AST interpretations in the filtered cohort.

Monthly volume

Included isolate records by specimen month.

Two things are always explicit on screen because they change every number: the **deduplication mode** (\$24) and the **date window**. A saved macro stores a whole filter set so the same analysis can be re-run next month and compared honestly.

9. One Health capture

AMR is not only a human-health problem. The **One Health** workbench captures veterinary, environmental, food-chain, antimicrobial-consumption and laboratory-quality events against the same facility registry, with its own operator accounts and its own governance.

The screenshot displays the ICMR AMRIT One Health workbench interface. The sidebar on the left contains navigation links for Workspace (Overview, Laboratories), Surveillance (Isolate records, Batch import, Analytics, Standards export, One Health), Configuration (Master Studio, Breakpoint Centre, Sync Centre, AI assistant), and Governance (Deployment & country, Audit trail). The main content area is titled 'One Health workbench' and includes a description: 'Capture and coordinate human, animal, environmental, food-chain, stewardship, IPC and genomic signals through auditable modules.' It features a grid of modules: Human laboratory quality, Antimicrobial consumption, Facility stewardship, IPC and HAI surveillance, Veterinary and fisheries, Food AMR and residues, Environmental AMR, and Reference laboratory and genomics. Below the modules, there are buttons for 'Aggregate data product', 'Refresh', 'Export', 'Enqueue aggregate', and 'Capture event'. A summary section shows counts for 'human_quality' (TOTAL: 1, VALIDATED: 1, DRAFTS: 0, FACILITIES: 1, OPEN ALERTS: 0). A table titled 'Human laboratory quality' lists events with columns: EVENT, SAMPLE / ACTIVITY, ORGANISM / INDICATOR, FACILITY, ACTOR, and QUALITY. The table shows one event: DEMO-LAB, contamination, contamination, DEMO-LAB, demo.admin, and VALID. A 'Priority alerts' section indicates 'No open alerts'.

10. Export

Format	What it is for
WHONET	Exchange with the WHONET/GLASS ecosystem
FHIR (Observation / MeasureReport)	Interoperable clinical and aggregate reporting
HL7 v2	Feeding a hospital information system
Measure bundle	Aggregate indicator reporting to a programme
CSV / JSON	Analysis in R, Python, Excel

The screenshot shows the 'Standards export' page in the ICMR AMRIT interface. The left sidebar contains navigation links for Workspace, Surveillance, Configuration, and Governance. The main content area is titled 'Standards export' under the 'INTEROPERABILITY' section. It includes a sub-section 'Export cohort' with date pickers for 'Period start' and 'Period end'. Below this are six export options, each with a description and a 'Save export' button: WHONET (.csv), FHIR R4 Bundle (.json), HL7 v2.5.1 (.hl7), FHIR MeasureReport (.json), Analysis CSV (.csv), and AMRIT JSON (.json). At the bottom, there is a section for 'Interoperability safeguards'.

An export is a deliberate act by the operator, and it may contain patient-level data — that is the difference between an export and federation (§28), and the audit log records both.

11. Join a central server (federation)

The screenshot shows the 'Sync Centre' page in the ICMR AMRIT interface. The left sidebar is the same as the previous screenshot. The main content area is titled 'Sync Centre' under the 'SECURE FEDERATION' section. It includes a sub-section 'Configure aggregate-only long-poll and WebSocket federation with encrypted tokens, explicit consent and live status.' with a 'Start sync' button. Below this are three status indicators: 'LONG-POLL WORKER' (Off), 'WEBSOCKET CHANNEL' (Off), and 'SITE CREDENTIAL' (Not Configured). The 'Server connection' section contains fields for 'Central server URL', 'Laboratory / site code', 'Authentication token', and 'Site token'. It also has input fields for 'Poll interval (seconds)' and 'Long-poll timeout (seconds)'. The 'Privacy contract' section lists several criteria: 'Aggregate-only responses', 'Allowlisted query types', 'Validated criteria', 'Audited requests', and 'Encrypted credentials'. The 'Optional location sharing' section has a toggle for 'I authorise site-coordinate sharing'. At the bottom, there is a section for 'Approved query allowlist'.

1. **Request access.** The desktop registers its lab code with the server and receives a one-time *pickup token*. Nothing is granted yet.
2. **An administrator approves** it in the portal, which creates the site and mints a *site token* delivered out of band.
3. **Collect the token.** The desktop redeems its pickup token for a bearer token it mints and stores in the operating system's credential vault. The server keeps only a hash.
4. **Allow-list the query types** this laboratory will answer. A query type not on the list is refused, logged and answered with an error.
5. **Start sync.** A long-poll worker asks for questions; a WebSocket connection lets the server nudge for an immediate answer. A four-hourly heartbeat reports presence.

Figure 2. Enrolment: nothing is granted by asking

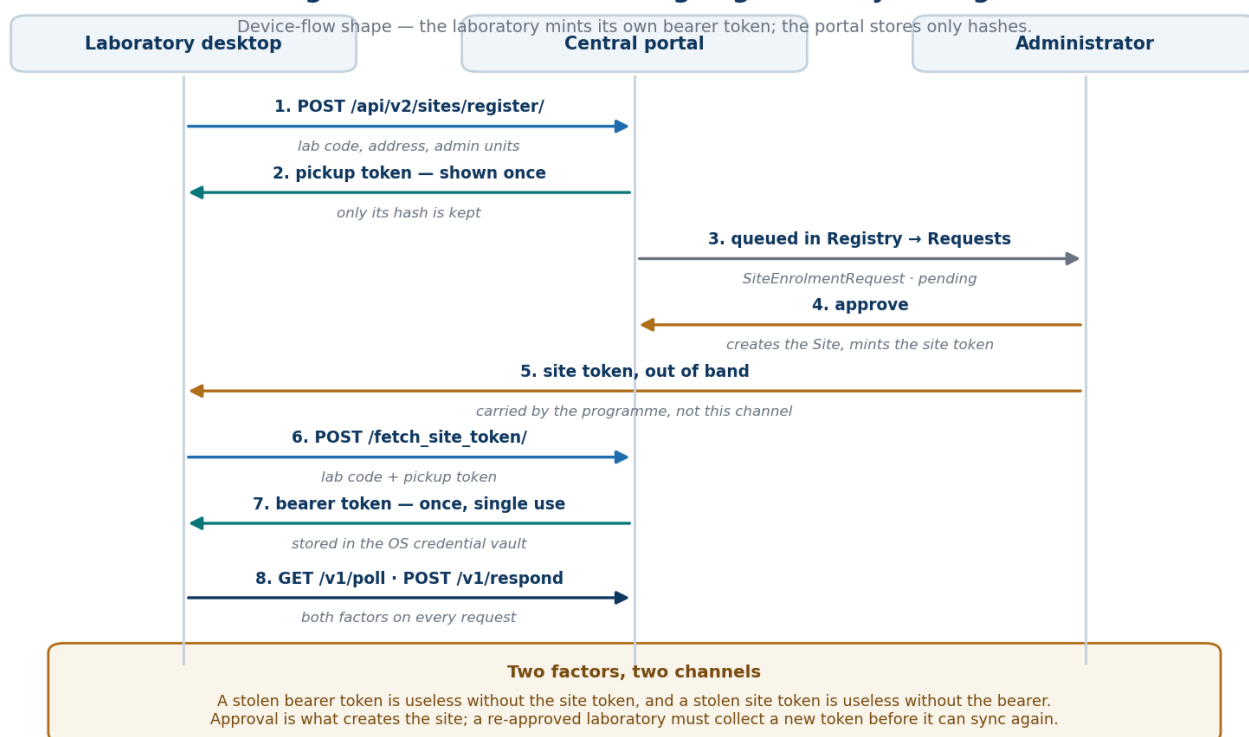


Figure 2 — enrolment and the two-token credential exchange.

12. Privacy, retention, audit

- **Retention.** The profile sets a retention period; expiry removes row-level data on a schedule, with a dry run that reports what would go first.
- **Erasure.** A named subject's rows can be erased on request, with the erasure itself audited.
- **Audit.** Every consequential act — saving a record, activating a breakpoint set, exporting a file, starting sync, answering a query — is written to an append-only local log.

**ICMR AMRIT**
AMR Intelligence Toolkit

WORKSPACE

Overview

Laboratories

SURVEILLANCE

Isolate records

Batch import

Analytics

Standards export

One Health

CONFIGURATION

Master Studio

Breakpoint Centre

Sync Centre

AI assistant

GOVERNANCE

Deployment & country

Audit trail

Local-first

Aggregate-only sync

ACTIVE LABORATORY

Demonstration Microbiology Laboratory · DEMO-LAB

SYNC OFFLINE

Settings

GOVERNANCE

Audit trail

Inspect traceable local operations across configuration, imports, records, exports, federation and optional AI services.

74

LOADED EVENTS

73

SUCCESSFUL

1

WARNINGS / ERRORS

All outcomes

74 events shown

TIME	OPERATION	SUMMARY	OUTCOME	DETAILS
14 Aug 2026, 11:56 am	ipc.error	llm.models: AI network access is off	ERROR	Inspect
14 Aug 2026, 11:56 am	breakpoints.eucast-update	950 EUCAST row(s) staged from network (15.0)	OK	Inspect
14 Aug 2026, 11:56 am	breakpoint.stage	EUCAST Breakpoint Tables v15.0: 950 row(s) staged inactive	OK	Inspect
14 Aug 2026, 11:55 am	one-health.capture	environment: 9f9fc90c-6fc5-4467-aa57-16af0ddfa08	OK	Inspect
14 Aug 2026, 11:55 am	one-health.capture	veterinary: 65b048c9-cd12-468f-9e65-ee3b9213a1ca	OK	Inspect
14 Aug 2026, 11:55 am	one-health.capture	amc: f143cca1-f573-4c01-9fb0-636afdbb8e3b	OK	Inspect
14 Aug 2026, 11:55 am	one-health.capture	human_quality: d12f25b0-8679-4da4-b7fa-5a95f558a84b	OK	Inspect
14 Aug 2026, 11:55 am	one-health.user.bootstrap	demo.admin: first administrator	OK	Inspect
14 Aug 2026, 11:55 am	isolate.save	Isolate 60	OK	Inspect

Operator manual

15 / 44

Part II — Central web portal

13. Deploy the server

The portal is Django with Postgres and Redis, shipped as a Docker Compose stack.

```
cd server/amrit_central_server
cp .env.example .env
# set at least: DJANGO_SECRET_KEY, DJANGO_ALLOWED_HOSTS, POSTGRES_PASSWORD,
#               AMRIT_COUNTRY_PROFILE (e.g. IN), AMRIT_ENROLMENT_SECRET
docker compose up -d --build
docker compose run --rm web python manage.py bootstrap # first administrator
```

The entrypoint migrates, collects static assets and — only when `AMRIT_SEED_DEMO=1` — seeds the demonstration pack for the configured country. A country with no pack is reported and skipped, never filled with another country's hospitals.

The stack publishes on loopback only. Nothing in it terminates TLS; put a reverse proxy in front before exposing it, because both operator sign-in and site enrolment carry credentials.

14. The site registry and enrolment

Registry → Sites lists every laboratory, its administrative unit, when it was last seen and whether it is online.

India · Indian Council of Medical Research

ICMR AMRIT — antimicrobial resistance surveillance

**ICMR AMRIT**
Indian Council of Medical Research

[Overview](#)[My Dashboard](#)[Action Plans](#)[Sites](#)[Map](#)[Queries](#)[Outbreaks](#)[NEW QUERY](#)[Audit](#)[Public](#)[Portal Admin](#)[SUPER ADMIN](#)[Sign out](#)

REGISTRY

Registered Sites

AMRIT lab installations in your scope.

SEARCH

lab code, name, geo...

STATUS

All

Filter

LAB	NAME	GEO	STATUS	LAST SEEN	LAST POLL	LAST RESPONSE	ACTIONS
PGIMER-CHD	PGIMER Chandigarh	India · PGIMER Chandigarh, Department of Microbiology, CHANDIGARH 600411, Chandigarh	idle	2026-08-14 05:55:03	2026-08-14 05:55:03	—	rotate edit rename remove
CMC-VLR	Christian Medical College Vellore	India · Christian Medical College Vellore, Department of Microbiology, VELLORE 600548, Tamil Nadu	idle	2026-08-14 05:54:53	2026-08-14 05:54:53	—	rotate edit rename remove
AIIMS-JDH	AIIMS Jodhpur	India · AIIMS Jodhpur, Department of Microbiology, JODHPUR 601507, Rajasthan	idle	2026-08-14 05:54:34	2026-08-14 05:54:34	—	rotate edit rename remove
KEM-MUM	KEM Hospital Mumbai	India · KEM Hospital Mumbai, Department of Microbiology, MUMBAI 600137, Maharashtra	idle	2026-08-14 05:53:48	2026-08-14 05:53:48	—	rotate edit rename remove
GMC-GHY	Gauhati Medical College	India · Gauhati Medical College, Department of Microbiology, KAMRUP METRO 601918, Assam	idle	2026-08-14 05:53:28	2026-08-14 05:53:28	—	rotate edit rename remove
AIIMS-BBSR	AIIMS Bhubaneswar	India · AIIMS Bhubaneswar, Department of Microbiology, KHORDHA 600274, Odisha	idle	2026-08-14 05:53:07	2026-08-14 05:53:07	—	rotate edit rename remove
AIIMS-DEL	AIIMS New Delhi	India · AIIMS New Delhi, Department of Microbiology, NEW DELHI 600000, Delhi	idle	2026-08-14 05:52:49	2026-08-14 05:52:49	—	rotate edit rename remove
AIIMS-PAT	AIIMS Patna	India · AIIMS Patna, Department of Microbiology, PATNA 601781, Bihar	idle	2026-08-14 05:52:03	2026-08-14 05:52:03	—	rotate edit rename remove
IPGMR-KOL	IPGMR & SSKM Kolkata	India · IPGMR & SSKM Kolkata, Department of Microbiology, KOLKATA 600000, West Bengal	idle	2026-08-14 05:51:42	2026-08-14 05:51:42	—	rotate edit rename remove

Registry → Requests is the enrolment queue: laboratories that have asked to join and are waiting for a decision. Approving creates the site and shows its site token exactly once.

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ICMR AMRIT — antimicrobial resistance surveillance

ICMR AMRIT
Indian Council of Medical Research

Overview My Dashboard Action Plans Sites Map Queries Outbreaks + NEW QUERY Audit Public Portal Admin SUPER ADMIN Sign out

REGISTRY

Registration requests

Laboratories asking to join. Nothing is registered, and no token exists, until one is approved here.

Registered sites

SHOW Awaiting decision Approved Rejected All

Clear collected (0) Clear all settled (0)

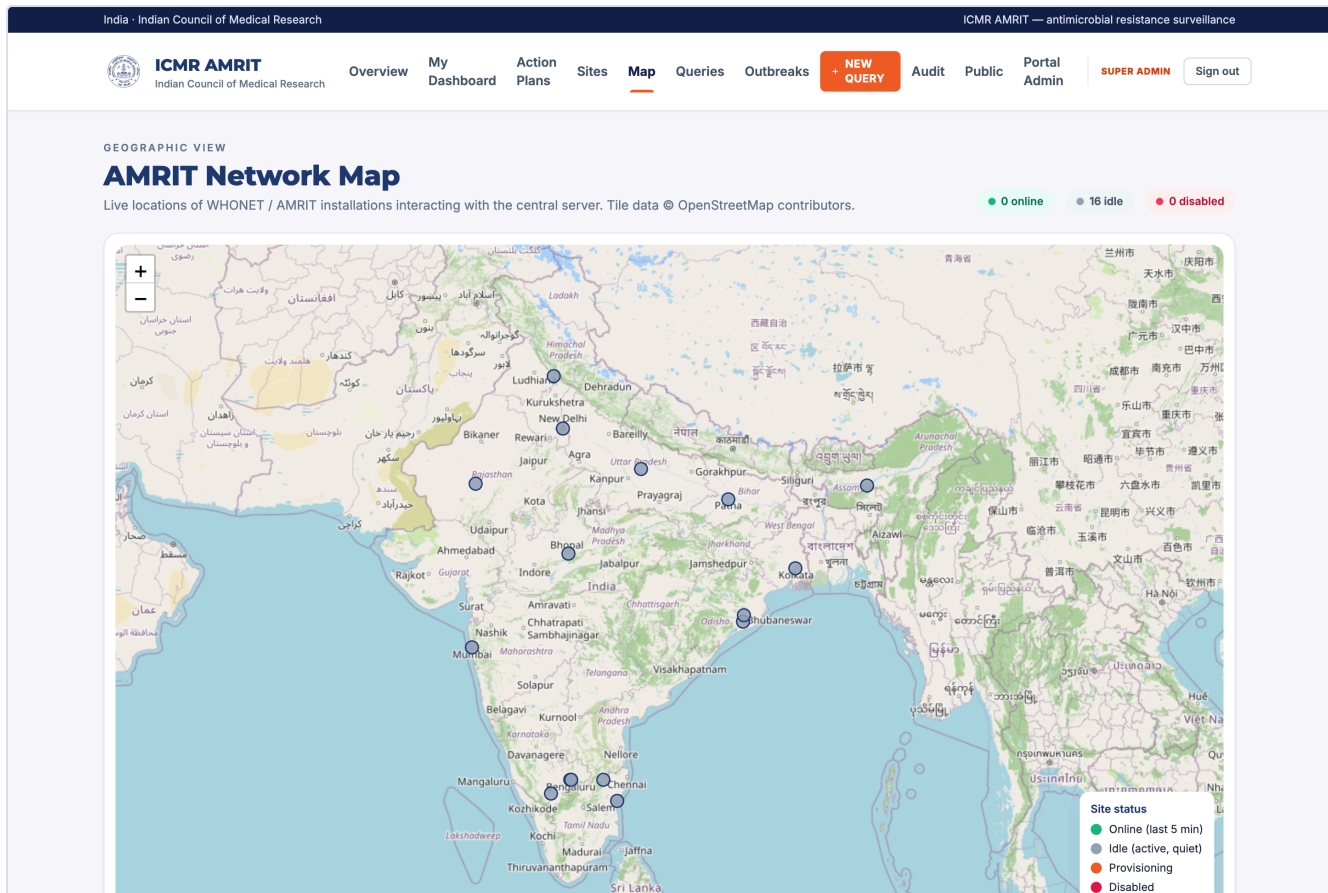
No requests with that status.

Indian Council of Medical Research
India - IND

ICMR AMRIT - central aggregation server
Aggregate, de-identified results only - k-anonymity floor enforced

Both credentials can be reset independently, and each reset stops that site syncing until the new value reaches it — which the screen says before the button is pressed.

Registry → Map places every consenting site.



15. Asking questions

Queries → New query composes a question and targets it: all sites, an administrative unit, or named laboratories.

India · Indian Council of Medical Research
ICMR AMRIT — antimicrobial resistance surveillance


ICMR AMRIT
Indian Council of Medical Research

[Overview](#)
[My Dashboard](#)
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[Sites](#)
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[Queries](#)
[Outbreaks](#)
[NEW QUERY](#)
[Audit](#)
[Public](#)
[Portal Admin](#)
[SUPER ADMIN](#)
[Sign out](#)

ALL SITES SCOPE · SUPER ADMIN

Live AMR data explorer

Aggregate, de-identified results from connected AMRIT desktop sites. Site access is enforced by role.

[Download filtered CSV](#)

Clinical & microbiology filters
Reset filters

Organism
Any organism

Specimen
Any specimen

Antibiotic
Any antibiotic

AST interpretation
S / I / R

From date
dd/mm/yyyy

To date
dd/mm/yyyy

Year
e.g. 2026

Patient sex
Any

Age band
Any

Patient setting
Any

Ward type
ICU, NICU, MED...

Patient type
Clinical, screening...

Sites in role scope
Select online Clear 0/16 selected

OFFLINE SITES

☐
AIIMS Bhopal

AIIMS-BPL · AIIMS Bhopal, Department of Microbiology, BHOPAL 601096, Madhya Pradesh

☐
AIIMS Bhubaneswar

AIIMS-BBSR · AIIMS Bhubaneswar, Department of Microbiology, KHANDA 600274, Odisha

☐
AIIMS Jodhpur

AIIMS-JDH · AIIMS Jodhpur, Department of Microbiology, JODHPUR 601507, Rajasthan

☐
AIIMS New Delhi

AIIMS-DEL · AIIMS New Delhi, Department of Microbiology, NEW DELHI 600000, Delhi

☐
AIIMS Patna

AIIMS-PAT · AIIMS Patna, Department of Microbiology, PATNA 601781, Bihar

☐
Bangalore Medical College

BMCR-BLR · Bangalore Medical College, Department of Microbiology, BENGALURU URBAN 600022, Karnataka

Run live analysis

Query type

Answer

isolate_count

How many isolates match the filters

organism_distribution

Counts by organism

specimen_distribution

Counts by specimen type

resistance_rate

Numerator, denominator and percentage for an organism/agent pair

measure_bundle

The same as a FHIR MeasureReport

cluster_scan

Daily case counts by site and phenotype, for outbreak detection

heartbeat

Presence only

Queries expire (`AMRIT_QUERY_TTL_SECONDS`), and each site's answer is recorded separately, so a partial answer is visibly partial.

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ICMR AMRIT — antimicrobial resistance surveillance


ICMR AMRIT
Indian Council of Medical Research

[Overview](#)
[My Dashboard](#)
[Action Plans](#)
[Sites](#)
[Map](#)
[Queries](#)
[Outbreaks](#)
[+ NEW QUERY](#)
[Audit](#)
[Public](#)
[Portal Admin](#)
[SUPER ADMIN](#)
[Sign out](#)

Aggregate Queries

Queries dispatched to AMRIT sites; aggregate, de-identified results only.

STATUS
All

TYPE
All

Filter

TITLE / TYPE	STATUS	ANTIBIOTIC	TARGETS	DISPATCH	RESULTS	CREATED	EXPIRES
No queries match.							

Indian Council of Medical Research
India · IND

ICMR AMRIT · central aggregation server
Aggregate, de-identified results only · k-anonymity floor enforced

16. Dashboards, by stakeholder

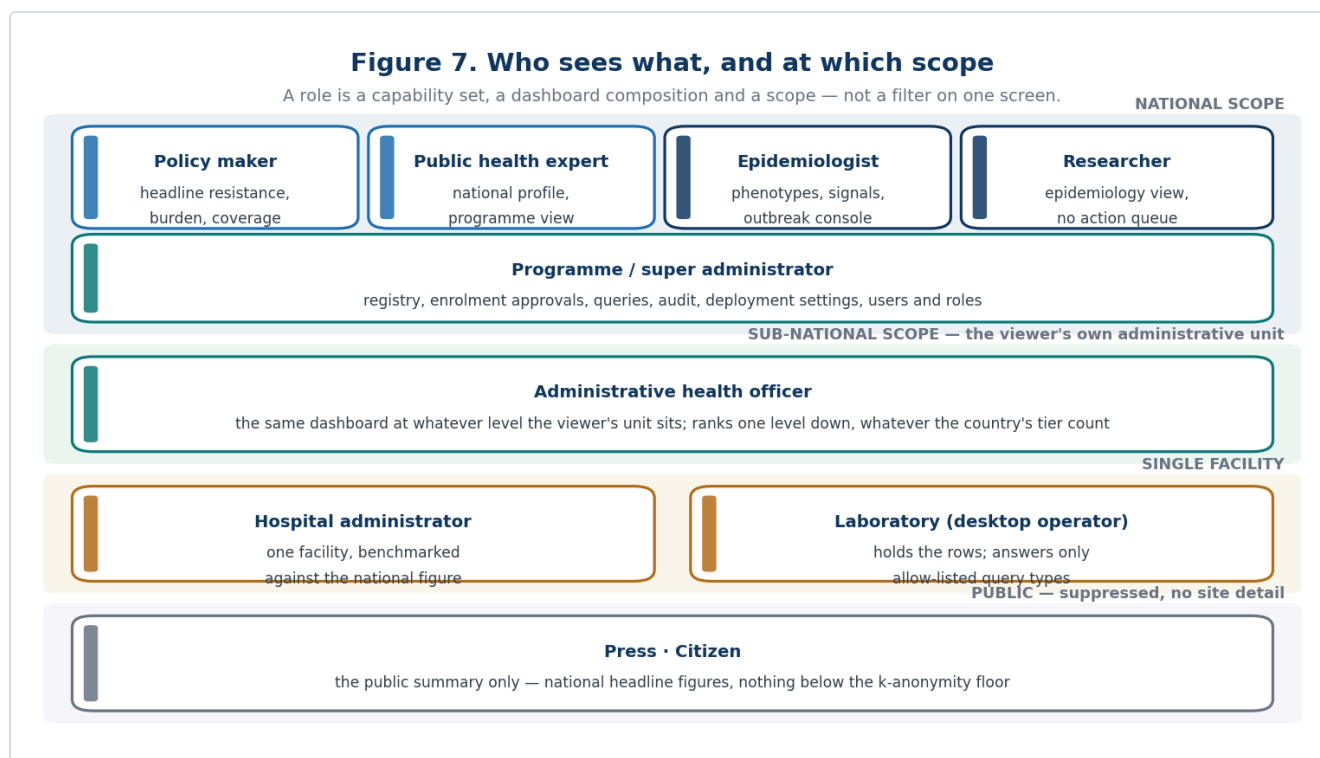


Figure 7 — who sees what, and at which scope.

A role is not a filter on one screen. Each stakeholder gets a different composition at a different scope, because the question each is asking is different.

Policy maker — national scope. Headline resistance, burden, coverage, trend.

Page not found (404)

Request Method: GET

Request URL: http://127.0.0.1:8011/dashboard/

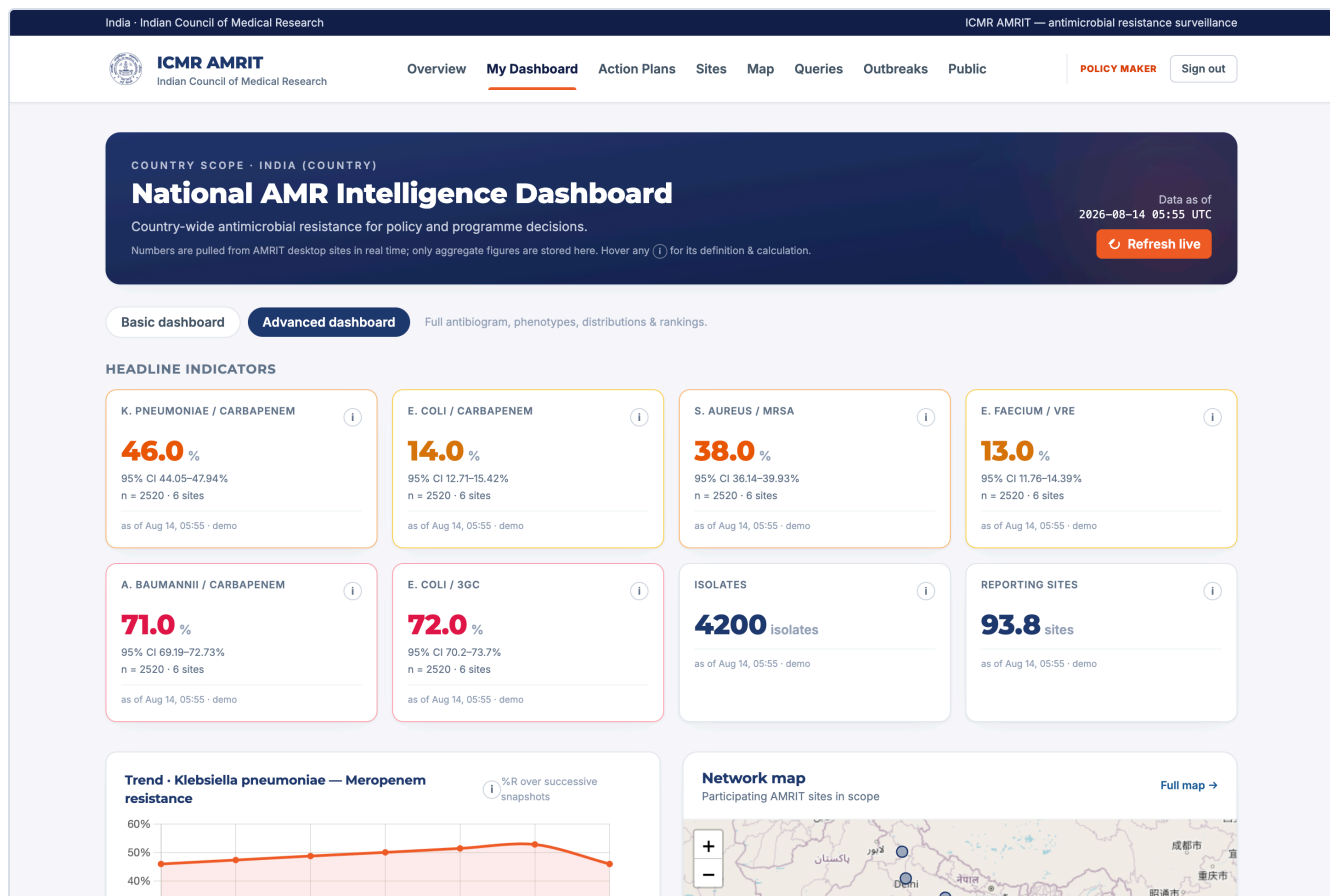
Using the URLconf defined in central.urls, Django tried these URL patterns, in this order:

```

1. [name='dashboard']
2. dashboard/licences/ [name='dashboard_licences']
3. dashboard/deployment/ [name='dashboard_deployment']
4. dashboard/deployment/save/ [name='dashboard_deployment_save']
5. dashboard/deployment/logo/ [name='dashboard_deployment_logo']
6. dashboard/deployment/reset/ [name='dashboard_deployment_reset']
7. dashboard/deployment/revert/ [name='dashboard_deployment_revert']
8. dashboard/deployment/export/ [name='dashboard_deployment_export']
9. dashboard/deployment/profile.json [name='dashboard_deployment_json']
10. dashboard/sites/ [name='dashboard_sites']
11. dashboard/sites/new/ [name='dashboard_site_create']
12. dashboard/sites/requests/ [name='dashboard_site_requests']
13. dashboard/sites/requests/clear/ [name='dashboard_site_requests_clear']
14. dashboard/sites/requests/<int:pk>/decide/ [name='dashboard_site_request_decide']
15. dashboard/sites/<str:lab_code>/edit/ [name='dashboard_site_edit']
16. dashboard/sites/<str:lab_code>/rename/ [name='dashboard_site_rename']
17. dashboard/sites/<str:lab_code>/delete/ [name='dashboard_site_delete']
18. dashboard/sites/<str:lab_code>/token/ [name='dashboard_site_token']
19. dashboard/sites/map/ [name='dashboard_map']
20. dashboard/sites/map.json [name='dashboard_map_json']
21. dashboard/queries/ [name='dashboard_queries']
22. dashboard/queries/new/ [name='query_new']
23. dashboard/queries/<uuid:pk>/ [name='query_detail']
24. dashboard/audit/ [name='dashboard_audit']
25. dashboard/public/ [name='public_summary']
26. dashboard/outbreaks/ [name='outbreak_dashboard']
27. portal-admin/ [name='portal_admin_home']
28. portal-admin/users/ [name='portal_admin_users']
29. portal-admin/users/new/ [name='portal_admin_user_new']
30. portal-admin/users/<int:pk>/ [name='portal_admin_user_edit']
31. portal-admin/users/<int:pk>/toggle/ [name='portal_admin_user_toggle']
32. portal-admin/roles/ [name='portal_admin_roles']
33. portal-admin/roles/new/ [name='portal_admin_role_new']
34. portal-admin/roles/<int:pk>/ [name='portal_admin_role_edit']
35. portal-admin/roles/<int:pk>/delete/ [name='portal_admin_role_delete']
36. accounts/login/ [name='login']
37. accounts/logout/ [name='logout']
38. dashboard/roles/ [name='dashboard_home']
39. dashboard/roles/<str:kind>/ [name='stakeholder_dashboard']
40. dashboard/roles/<str:kind>/refresh-live/ [name='dashboard_refresh_live']
41. actions/ [name='action_inbox']
42. actions/tracking/ [name='action_tracking']
43. actions/new/ [name='action_plan_new']
44. actions/<int:pk>/ [name='action_plan_detail']
45. health/ [name='health']
46. admin/
47. v1/
48. api/v1/sites/
49. api/v1/queries/
50. api/v1/analytics/
51. api/v1/ecosystem/
52. api/v1/schema/ [name='schema']
53. api/v1/docs/ [name='docs']
54. create_labcode/ [name='create_labcode']
55. create_labcode/<str:lab_code>/ [name='create_labcode_detail']

```

The advanced section opens the full resistance matrix and phenotype panels.



Epidemiologist — national scope with signals. Priority-pathogen phenotypes (CRE, MDR), anomaly signals and the outbreak console.

Page not found (404)

Request Method: GET

Request URL: http://127.0.0.1:8011/dashboard/

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4. dashboard/deployment/save/ [name='dashboard_deployment_save']
5. dashboard/deployment/logo/ [name='dashboard_deployment_logo']
6. dashboard/deployment/reset/ [name='dashboard_deployment_reset']
7. dashboard/deployment/revert/ [name='dashboard_deployment_revert']
8. dashboard/deployment/export/ [name='dashboard_deployment_export']
9. dashboard/deployment/profile.json [name='dashboard_deployment_json']
10. dashboard/sites/ [name='dashboard_sites']
11. dashboard/sites/new/ [name='dashboard_site_create']
12. dashboard/sites/requests/ [name='dashboard_site_requests']
13. dashboard/sites/requests/clear/ [name='dashboard_site_requests_clear']
14. dashboard/sites/requests/<int:pk>/decide/ [name='dashboard_site_request_decide']
15. dashboard/sites/<str:lab_code>/edit/ [name='dashboard_site_edit']
16. dashboard/sites/<str:lab_code>/rename/ [name='dashboard_site_rename']
17. dashboard/sites/<str:lab_code>/delete/ [name='dashboard_site_delete']
18. dashboard/sites/<str:lab_code>/token/ [name='dashboard_site_token']
19. dashboard/sites/map/ [name='dashboard_map']
20. dashboard/sites/map.json [name='dashboard_map_json']
21. dashboard/queries/ [name='dashboard_queries']
22. dashboard/queries/new/ [name='query_new']
23. dashboard/queries/<uuid:pk>/ [name='query_detail']
24. dashboard/audit/ [name='dashboard_audit']
25. dashboard/public/ [name='public_summary']
26. dashboard/outbreaks/ [name='outbreak_dashboard']
27. portal-admin/ [name='portal_admin_home']
28. portal-admin/users/ [name='portal_admin_users']
29. portal-admin/users/new/ [name='portal_admin_user_new']
30. portal-admin/users/<int:pk>/ [name='portal_admin_user_edit']
31. portal-admin/users/<int:pk>/toggle/ [name='portal_admin_user_toggle']
32. portal-admin/roles/ [name='portal_admin_roles']
33. portal-admin/roles/new/ [name='portal_admin_role_new']
34. portal-admin/roles/<int:pk>/ [name='portal_admin_role_edit']
35. portal-admin/roles/<int:pk>/delete/ [name='portal_admin_role_delete']
36. accounts/login/ [name='login']
37. accounts/logout/ [name='logout']
38. dashboard/roles/ [name='dashboard_home']
39. dashboard/roles/<str:kind>/ [name='stakeholder_dashboard']
40. dashboard/roles/<str:kind>/refresh-live/ [name='dashboard_refresh_live']
41. actions/ [name='action_inbox']
42. actions/tracking/ [name='action_tracking']
43. actions/new/ [name='action_plan_new']
44. actions/<int:pk>/ [name='action_plan_detail']
45. health/ [name='health']
46. admin/
47. v1/
48. api/v1/sites/
49. api/v1/queries/
50. api/v1/analytics/
51. api/v1/ecosystem/
52. api/v1/schema/ [name='schema']
53. api/v1/docs/ [name='docs']
54. create_labcode/ [name='create_labcode']
```

Researcher. The epidemiology composition without the operational action queue.

Page not found (404)

Request Method: GET

Request URL: http://127.0.0.1:8011/dashboard/

Using the URLconf defined in central.urls, Django tried these URL patterns, in this order:

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4. dashboard/deployment/save/ [name='dashboard_deployment_save']
5. dashboard/deployment/logo/ [name='dashboard_deployment_logo']
6. dashboard/deployment/reset/ [name='dashboard_deployment_reset']
7. dashboard/deployment/revert/ [name='dashboard_deployment_revert']
8. dashboard/deployment/export/ [name='dashboard_deployment_export']
9. dashboard/deployment/profile.json [name='dashboard_deployment_json']
10. dashboard/sites/ [name='dashboard_sites']
11. dashboard/sites/new/ [name='dashboard_site_create']
12. dashboard/sites/requests/ [name='dashboard_site_requests']
13. dashboard/sites/requests/clear/ [name='dashboard_site_requests_clear']
14. dashboard/sites/requests/<int:pk>/decide/ [name='dashboard_site_request_decide']
15. dashboard/sites/<str:lab_code>/edit/ [name='dashboard_site_edit']
16. dashboard/sites/<str:lab_code>/rename/ [name='dashboard_site_rename']
17. dashboard/sites/<str:lab_code>/delete/ [name='dashboard_site_delete']
18. dashboard/sites/<str:lab_code>/token/ [name='dashboard_site_token']
19. dashboard/sites/map/ [name='dashboard_map']
20. dashboard/sites/map.json [name='dashboard_map_json']
21. dashboard/queries/ [name='dashboard_queries']
22. dashboard/queries/new/ [name='query_new']
23. dashboard/queries/<uuid:pk>/ [name='query_detail']
24. dashboard/audit/ [name='dashboard_audit']
25. dashboard/public/ [name='public_summary']
26. dashboard/outbreaks/ [name='outbreak_dashboard']
27. portal-admin/ [name='portal_admin_home']
28. portal-admin/users/ [name='portal_admin_users']
29. portal-admin/users/new/ [name='portal_admin_user_new']
30. portal-admin/users/<int:pk>/ [name='portal_admin_user_edit']
31. portal-admin/users/<int:pk>/toggle/ [name='portal_admin_user_toggle']
32. portal-admin/roles/ [name='portal_admin_roles']
33. portal-admin/roles/new/ [name='portal_admin_role_new']
34. portal-admin/roles/<int:pk>/ [name='portal_admin_role_edit']
35. portal-admin/roles/<int:pk>/delete/ [name='portal_admin_role_delete']
36. accounts/login/ [name='login']
37. accounts/logout/ [name='logout']
38. dashboard/roles/ [name='dashboard_home']
39. dashboard/roles/<str:kind>/ [name='stakeholder_dashboard']
40. dashboard/roles/<str:kind>/refresh-live/ [name='dashboard_refresh_live']
41. actions/ [name='action_inbox']
42. actions/tracking/ [name='action_tracking']
43. actions/new/ [name='action_plan_new']
44. actions/<int:pk>/ [name='action_plan_detail']
45. health/ [name='health']
46. admin/
47. v1/
48. api/v1/sites/
49. api/v1/queries/
50. api/v1/analytics/
51. api/v1/ecosystem/
52. api/v1/schema/ [name='schema']
53. api/v1/docs/ [name='docs']
54. create_labcode/ [name='create_labcode']
```

Public health expert — national scope.

Page not found (404)

Request Method: GET

Request URL: http://127.0.0.1:8011/dashboard/

Using the URLconf defined in central.urls, Django tried these URL patterns, in this order:

```

1. [name='dashboard']
2. dashboard/licences/ [name='dashboard_licences']
3. dashboard/deployment/ [name='dashboard_deployment']
4. dashboard/deployment/save/ [name='dashboard_deployment_save']
5. dashboard/deployment/logo/ [name='dashboard_deployment_logo']
6. dashboard/deployment/reset/ [name='dashboard_deployment_reset']
7. dashboard/deployment/revert/ [name='dashboard_deployment_revert']
8. dashboard/deployment/export/ [name='dashboard_deployment_export']
9. dashboard/deployment/profile.json [name='dashboard_deployment_json']
10. dashboard/sites/ [name='dashboard_sites']
11. dashboard/sites/new/ [name='dashboard_site_create']
12. dashboard/sites/requests/ [name='dashboard_site_requests']
13. dashboard/sites/requests/clear/ [name='dashboard_site_requests_clear']
14. dashboard/sites/requests/<int:pk>/decide/ [name='dashboard_site_request_decide']
15. dashboard/sites/<str:lab_code>/edit/ [name='dashboard_site_edit']
16. dashboard/sites/<str:lab_code>/rename/ [name='dashboard_site_rename']
17. dashboard/sites/<str:lab_code>/delete/ [name='dashboard_site_delete']
18. dashboard/sites/<str:lab_code>/token/ [name='dashboard_site_token']
19. dashboard/sites/map/ [name='dashboard_map']
20. dashboard/sites/map.json [name='dashboard_map_json']
21. dashboard/queries/ [name='dashboard_queries']
22. dashboard/queries/new/ [name='query_new']
23. dashboard/queries/<uuid:pk>/ [name='query_detail']
24. dashboard/audit/ [name='dashboard_audit']
25. dashboard/public/ [name='public_summary']
26. dashboard/outbreaks/ [name='outbreak_dashboard']
27. portal-admin/ [name='portal_admin_home']
28. portal-admin/users/ [name='portal_admin_users']
29. portal-admin/users/new/ [name='portal_admin_user_new']
30. portal-admin/users/<int:pk>/ [name='portal_admin_user_edit']
31. portal-admin/users/<int:pk>/toggle/ [name='portal_admin_user_toggle']
32. portal-admin/roles/ [name='portal_admin_roles']
33. portal-admin/roles/new/ [name='portal_admin_role_new']
34. portal-admin/roles/<int:pk>/ [name='portal_admin_role_edit']
35. portal-admin/roles/<int:pk>/delete/ [name='portal_admin_role_delete']
36. accounts/login/ [name='login']
37. accounts/logout/ [name='logout']
38. dashboard/roles/ [name='dashboard_home']
39. dashboard/roles/<str:kind>/ [name='stakeholder_dashboard']
40. dashboard/roles/<str:kind>/refresh-live/ [name='dashboard_refresh_live']
41. actions/ [name='action_inbox']
42. actions/tracking/ [name='action_tracking']
43. actions/new/ [name='action_plan_new']
44. actions/<int:pk>/ [name='action_plan_detail']
45. health/ [name='health']
46. admin/
47. v1/
48. api/v1/sites/
49. api/v1/queries/
50. api/v1/analytics/
51. api/v1/ecosystem/
52. api/v1/schema/ [name='schema']
53. api/v1/docs/ [name='docs']
54. create_labcode/ [name='create_labcode'] ...

```

Administrative health officer — sub-national scope. The same dashboard at whatever level the viewer's own administrative unit sits, ranking one level down. A three-tier country needs three *units*, not three dashboards.

Page not found (404)

Request Method: GET

Request URL: http://127.0.0.1:8011/dashboard/

Using the URLconf defined in central.urls, Django tried these URL patterns, in this order:

```

1. [name='dashboard']
2. dashboard/licences/ [name='dashboard_licences']
3. dashboard/deployment/ [name='dashboard_deployment']
4. dashboard/deployment/save/ [name='dashboard_deployment_save']
5. dashboard/deployment/logo/ [name='dashboard_deployment_logo']
6. dashboard/deployment/reset/ [name='dashboard_deployment_reset']
7. dashboard/deployment/revert/ [name='dashboard_deployment_revert']
8. dashboard/deployment/export/ [name='dashboard_deployment_export']
9. dashboard/deployment/profile.json [name='dashboard_deployment_json']
10. dashboard/sites/ [name='dashboard_sites']
11. dashboard/sites/new/ [name='dashboard_site_create']
12. dashboard/sites/requests/ [name='dashboard_site_requests']
13. dashboard/sites/requests/clear/ [name='dashboard_site_requests_clear']
14. dashboard/sites/requests/<int:pk>/decide/ [name='dashboard_site_request_decide']
15. dashboard/sites/<str:lab_code>/edit/ [name='dashboard_site_edit']
16. dashboard/sites/<str:lab_code>/rename/ [name='dashboard_site_rename']
17. dashboard/sites/<str:lab_code>/delete/ [name='dashboard_site_delete']
18. dashboard/sites/<str:lab_code>/token/ [name='dashboard_site_token']
19. dashboard/sites/map/ [name='dashboard_map']
20. dashboard/sites/map.json [name='dashboard_map_json']
21. dashboard/queries/ [name='dashboard_queries']
22. dashboard/queries/new/ [name='query_new']
23. dashboard/queries/<uuid:pk>/ [name='query_detail']
24. dashboard/audit/ [name='dashboard_audit']
25. dashboard/public/ [name='public_summary']
26. dashboard/outbreaks/ [name='outbreak_dashboard']
27. portal-admin/ [name='portal_admin_home']
28. portal-admin/users/ [name='portal_admin_users']
29. portal-admin/users/new/ [name='portal_admin_user_new']
30. portal-admin/users/<int:pk>/ [name='portal_admin_user_edit']
31. portal-admin/users/<int:pk>/toggle/ [name='portal_admin_user_toggle']
32. portal-admin/roles/ [name='portal_admin_roles']
33. portal-admin/roles/new/ [name='portal_admin_role_new']
34. portal-admin/roles/<int:pk>/ [name='portal_admin_role_edit']
35. portal-admin/roles/<int:pk>/delete/ [name='portal_admin_role_delete']
36. accounts/login/ [name='login']
37. accounts/logout/ [name='logout']
38. dashboard/roles/ [name='dashboard_home']
39. dashboard/roles/<str:kind>/ [name='stakeholder_dashboard']
40. dashboard/roles/<str:kind>/refresh-live/ [name='dashboard_refresh_live']
41. actions/ [name='action_inbox']
42. actions/tracking/ [name='action_tracking']
43. actions/new/ [name='action_plan_new']
44. actions/<int:pk>/ [name='action_plan_detail']
45. health/ [name='health']
46. admin/
47. v1/
48. api/v1/sites/
49. api/v1/queries/
50. api/v1/analytics/
51. api/v1/ecosystem/
52. api/v1/schema/ [name='schema']
53. api/v1/docs/ [name='docs']
54. create_labcode/ [name='create_labcode']

```

Hospital administrator — single-facility scope. One facility's profile, benchmarked against the national figure.

Page not found (404)

Request Method: GET

Request URL: http://127.0.0.1:8011/dashboard/

Using the URLconf defined in central.urls, Django tried these URL patterns, in this order:

```
1. [name='dashboard']
2. dashboard/licences/ [name='dashboard_licences']
3. dashboard/deployment/ [name='dashboard_deployment']
4. dashboard/deployment/save/ [name='dashboard_deployment_save']
5. dashboard/deployment/logo/ [name='dashboard_deployment_logo']
6. dashboard/deployment/reset/ [name='dashboard_deployment_reset']
7. dashboard/deployment/revert/ [name='dashboard_deployment_revert']
8. dashboard/deployment/export/ [name='dashboard_deployment_export']
9. dashboard/deployment/profile.json [name='dashboard_deployment_json']
10. dashboard/sites/ [name='dashboard_sites']
11. dashboard/sites/new/ [name='dashboard_site_create']
12. dashboard/sites/requests/ [name='dashboard_site_requests']
13. dashboard/sites/requests/clear/ [name='dashboard_site_requests_clear']
14. dashboard/sites/requests/<int:pk>/decide/ [name='dashboard_site_request_decide']
15. dashboard/sites/<str:lab_code>/edit/ [name='dashboard_site_edit']
16. dashboard/sites/<str:lab_code>/rename/ [name='dashboard_site_rename']
17. dashboard/sites/<str:lab_code>/delete/ [name='dashboard_site_delete']
18. dashboard/sites/<str:lab_code>/token/ [name='dashboard_site_token']
19. dashboard/sites/map/ [name='dashboard_map']
20. dashboard/sites/map.json [name='dashboard_map_json']
21. dashboard/queries/ [name='dashboard_queries']
22. dashboard/queries/new/ [name='query_new']
23. dashboard/queries/<uuid:pk>/ [name='query_detail']
24. dashboard/audit/ [name='dashboard_audit']
25. dashboard/public/ [name='public_summary']
26. dashboard/outbreaks/ [name='outbreak_dashboard']
27. portal-admin/ [name='portal_admin_home']
28. portal-admin/users/ [name='portal_admin_users']
29. portal-admin/users/new/ [name='portal_admin_user_new']
30. portal-admin/users/<int:pk>/ [name='portal_admin_user_edit']
31. portal-admin/users/<int:pk>/toggle/ [name='portal_admin_user_toggle']
32. portal-admin/roles/ [name='portal_admin_roles']
33. portal-admin/roles/new/ [name='portal_admin_role_new']
34. portal-admin/roles/<int:pk>/ [name='portal_admin_role_edit']
35. portal-admin/roles/<int:pk>/delete/ [name='portal_admin_role_delete']
36. accounts/login/ [name='login']
37. accounts/logout/ [name='logout']
38. dashboard/roles/ [name='dashboard_home']
39. dashboard/roles/<str:kind>/ [name='stakeholder_dashboard']
40. dashboard/roles/<str:kind>/refresh-live/ [name='dashboard_refresh_live']
41. actions/ [name='action_inbox']
42. actions/tracking/ [name='action_tracking']
43. actions/new/ [name='action_plan_new']
44. actions/<int:pk>/ [name='action_plan_detail']
45. health/ [name='health']
46. admin/
47. v1/
48. api/v1/sites/
49. api/v1/queries/
50. api/v1/analytics/
51. api/v1/ecosystem/
52. api/v1/schema/ [name='schema']
53. api/v1/docs/ [name='docs']
54. create_labcode/ [name='create_labcode']
```

Press and citizen. The public summary only (\$19).

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Overview **Public**

PRESS / MEDIA Sign out

PUBLIC SUMMARY

Antimicrobial resistance surveillance, India

Aggregate, de-identified surveillance metrics from connected laboratories across India.
No patient-level data is exposed.

Figures covering fewer than 5 isolates are suppressed rather than published.

ACTIVE LABS
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ONLINE NOW
0

STATES & UTS COVERED
0

COMPLETED STUDIES
0

Recent completed studies

TITLE	TYPE	ANTIBIOTIC	COMPLETED
No completed studies yet.			

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Aggregate, de-identified results only · k-anonymity floor enforced

Refresh live dispatches a fresh batch of queries to the in-scope sites, waits for the answers, and recomputes the snapshots for that scope (Figure 3). Historical results are never folded into a new refresh's numbers.

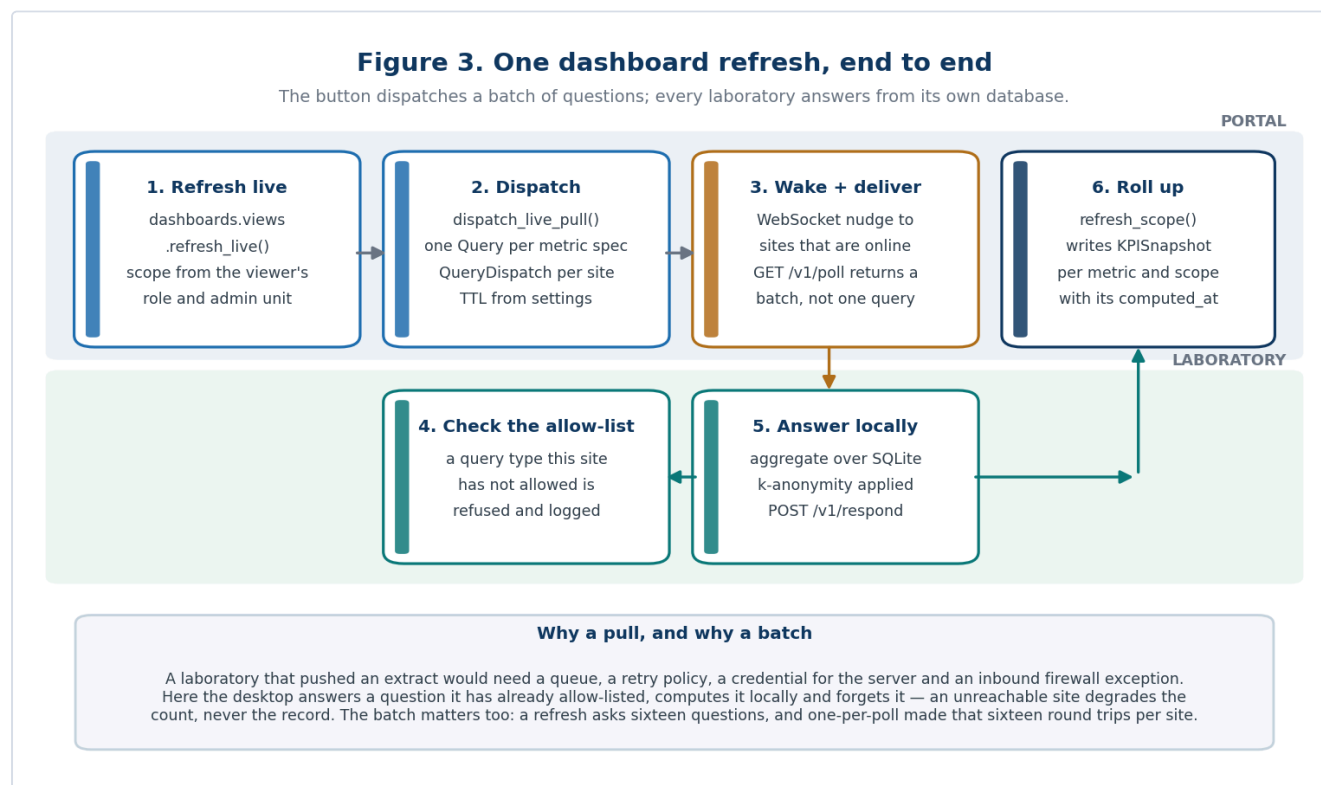


Figure 3 — one dashboard refresh, end to end.

17. The outbreak console

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ICMR AMRIT — antimicrobial resistance surveillance

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Indian Council of Medical Research

Overview My Dashboard Action Plans Sites Map Queries **Outbreaks** + NEW QUERY Audit Public Portal Admin SUPER ADMIN Sign out

RESTRICTED EPIDEMIOLOGY WORKSPACE

Organism & resistance outbreak detection

WHONET-compatible Kulldorff space-time permutation scan over aggregate daily counts. Expected values condition on site and time marginals; Monte Carlo maximum-statistic testing controls overlapping windows and signals.

Statistical signal ≠ confirmed outbreak. Infection-prevention staff must review patient overlap locally, specimen and testing changes, contamination, epidemiological links, typing/WGS, and plausible transmission before action or external reporting.

1 • Collect fresh aggregate inputs

Each site returns daily organism and recorded-resistant (R) phenotype counts after rolling patient-organism de-duplication. No patient, specimen, ward, or free-text identifier leaves the site.

Start date: 15/08/2025

End date: 14/08/2026

Repeat-isolate interval: 30 days; WHONET-style

Queue for visible sites

2 • Configure statistical scan

Input query: No outbreak queries

Analysis type: Prospective - ongoing only

Target: Organisms + resistance

Baseline days: 365

Max cluster days: 60

Minimum cases: 3

Monte Carlo replications: 999 - standard

Alert recurrence threshold: 365 days under null

Run scan

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The console runs the space-time permutation scan (§26) over `cluster_scan` aggregates and lists clusters with their window, sites, observed and expected counts, p-value and recurrence interval. Settings — baseline, maximum cluster length, minimum cases, permutations — are on the screen, because a signal without its parameters is not interpretable.

18. From a signal to an action plan

Threshold rules turn a metric crossing into a **draft** action plan — never an automatic instruction. A person edits, assigns and schedules it; tracking records what was done.

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Action plans

Draft plans raised when indicators breach limits, plus plans addressed to your role & scope.

Open (6)

Draft (6)

All (6)

Tracking board

+ New plan

PLAN	SCOPE	SEVERITY	STATUS	PROGRESS	RAISED
Facility action: carbapenem-resistant K. pneumoniae at 45.1% ⓘ auto-raised · res_kpn_carbapenem = 45.1% · for hospital_admin	Site / facility · NIMHANS-BLR	High	Draft	0/4	2026-08-14
07: carbapenem-resistant K. pneumoniae at 52.9% ⓘ auto-raised · res_kpn_carbapenem = 52.9% · for admin_officer	Administrative level 1 · 07	High	Draft	0/4	2026-08-14
27: carbapenem-resistant K. pneumoniae at 49.7% ⓘ auto-raised · res_kpn_carbapenem = 49.7% · for admin_officer	Administrative level 1 · 27	High	Draft	0/4	2026-08-14
29: carbapenem-resistant K. pneumoniae at 42.3% ⓘ auto-raised · res_kpn_carbapenem = 42.3% · for admin_officer	Administrative level 1 · 29	High	Draft	0/4	2026-08-14
National watch: MRSA at 38.0% ⓘ auto-raised · res_sau_mrsa = 38.0% · for policy_maker	Country	High	Draft	0/3	2026-08-14
National alert: carbapenem-resistant K. pneumoniae at 46.0% ⓘ auto-raised · res_kpn_carbapenem = 46.0% · for policy_maker	Country	Critical	Draft	0/4	2026-08-14

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Action tracking board

Status of every action plan in your scope.

Back to inbox

TOTAL PLANS
6

OPEN
6

DRAFT
6

OVERDUE
0

By status

Draft	6
Issued	0
In_progress	0
Completed	0
Closed	0
Dismissed	0

By severity

Info	0
Moderate	0
High	5
Critical	1

Overdue plans

Nothing overdue. 🚫

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Operator manual

29 / 44

19. The public view

A deliberately reduced summary for press and citizens: national headline figures, no site-level detail, everything already suppressed below the k-anonymity floor.

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PUBLIC SUMMARY

Antimicrobial resistance surveillance, India

Aggregate, de-identified surveillance metrics from connected laboratories across India.
No patient-level data is exposed.

Figures covering fewer than 5 isolates are suppressed rather than published.

ACTIVE LABS

16

ONLINE NOW

0

STATES & UTS COVERED

0

COMPLETED STUDIES

0

Recent completed studies

TITLE	TYPE	ANTIBIOTIC	COMPLETED
No completed studies yet.			

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20. Audit and governance

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Audit Log

Server-side mirror of every site poll, dispatch, and response.

ACTION

LAB CODE

All ▾

e.g. LAB-001

Filter

TIME	LAB	ACTION	QUERY	DETAIL
No audit events match.				

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Every dispatch, answer, approval, token reset, rename and deletion is recorded with actor, time and detail. **Licences** lists every bundled dataset and its terms.

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Open data and licences

The reference datasets this software bundles, and the terms they come under. None of it is patient data.

Requires action by this deployment

SNOMED CT concept codes — Requires a licence. Free to use in a SNOMED International Member country or territory; an affiliate licence is required elsewhere. This software grants no SNOMED CT licence and a deployment must establish its own position. [Licensing details](#)

ISO 3166-2 country subdivisions — Used under the deploying organisation's ISO licence. Subdivision codes and names are reproduced from the standard, so a deployment redistributing this software needs its own ISO licence. [Licensing details](#)

Dataset	Source	Terms	Bundled
WHONET reference catalogue Organisms, antimicrobials, expert rules, expected resistance, MIC panels, code values and field definitions.	WHONET / WHO Collaborating Centre for Surveillance of Antimicrobial Resistance	Distributed for antimicrobial resistance surveillance use. Terms	Yes
SNOMED CT concept codes Concept codes carried on organism rows. Supplementary: WHONET codes are the primary organism identifier, so the application functions with or without SNOMED in use.	SNOMED International	Requires a licence. Free to use in a SNOMED International Member country or territory; an affiliate licence is required elsewhere. This software grants no SNOMED CT licence and a deployment must establish its own position. Terms	Yes
LOINC codes Susceptibility and measurement codes on antimicrobial rows.	Regenstrief Institute	Free to use under the LOINC licence, with attribution. Terms	Yes
ATC codes and defined daily doses Used for antimicrobial consumption measures.	WHO Collaborating Centre for Drug Statistics Methodology	Free for non-commercial use; redistribution of the complete index is restricted. Terms	Yes

21. Portal administration

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ACCESS CONTROL

Portal administration

Manage accounts, passwords, role definitions, capabilities, dashboards, and geographic scope.

Users

11

Active

11

Roles

10

Sites

16

Users

Create accounts, assign roles/sites, reset passwords, activate or deactivate.

Roles & permissions

Create roles and define navigation, actions, dashboard, and data scope.

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Users and roles are managed in the portal itself. A role is a set of capabilities plus a dashboard kind plus a scope; a deployment can rename roles and change what each may see without a code change.

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← Administration

Users

Add user

USER	ROLE	SCOPE	STATUS	ACTIONS
citizen citizen@amrit.demo	Citizen	Country-wide / none	Active	Edit/reset Deactivate
district_officer district_officer@amrit.demo	District Health Officer	Country-wide / none	Active	Edit/reset Deactivate
epidemiologist epidemiologist@amrit.demo	Epidemiologist	Country-wide / none	Active	Edit/reset Deactivate
hospital_admin hospital_admin@amrit.demo	Hospital Administrator	NIMHANS-BLR	Active	Edit/reset Deactivate
policy_maker policy_maker@amrit.demo	Policy Maker	Country-wide / none	Active	Edit/reset Deactivate
press press@amrit.demo	Press / Media	Country-wide / none	Active	Edit/reset Deactivate
programme_admin programme_admin@amrit.demo	Programme Administrator	Country-wide / none	Active	Edit/reset Deactivate
public_health public_health@amrit.demo	Public Health Expert	Country-wide / none	Active	Edit/reset Deactivate
researcher researcher@amrit.demo	Researcher	Country-wide / none	Active	Edit/reset Deactivate
state_officer state_officer@amrit.demo	District Health Officer	Country-wide / none	Active	Edit/reset Deactivate
superadmin superadmin@amrit.demo	Super Admin	Country-wide / none	Active	Edit/reset Deactivate

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Roles & permissions

Add role

Citizen
citizen · system

Active

No description.

Dashboards: Operations / public only · Scope: No sites · Users: 1

view_public_summary

Edit

District Health Officer
admin_officer · system

Active

No description.

Dashboards: Administrative unit · Scope: Profile administrative unit · Users: 2

view_dashboard

view_scoped_sites

view_map

run_query

view_queries

view_public_summary

view_basic_dashboard

view_advanced_dashboard

manage_action_plans

track_action_points

Edit

Epidemiologist
epidemiologist · system

Active

No description.

Dashboards: Epidemiology · Scope: All sites · Users: 1

view_dashboard

view_all_sites

view_map

run_query

view_queries

view_audit

view_public_summary

view_basic_dashboard

view_advanced_dashboard

manage_action_plans

track_action_points

Edit

Hospital Administrator
hospital_admin · system

Active

No description.

Dashboards: Hospital · Scope: Assigned site · Users: 1

view_dashboard

view_own_site

run_query

view_queries

view_public_summary

view_basic_dashboard

view_advanced_dashboard

track_action_points

Edit

Policy Maker
policy_maker · system

Active

No description.

Dashboards: Country-wide · Scope: All sites · Users: 1

view_dashboard

view_all_sites

view_map

view_queries

view_public_summary

Press / Media
press · system

Active

No description.

Dashboards: Operations / public only · Scope: No sites · Users: 1

view_dashboard

view_public_summary

Part III — Methods and rationale

This part states, for each analytical choice: what the method is, what the alternatives were, and why this one was taken. It is the material a paper would cite.

22. Data model and standards

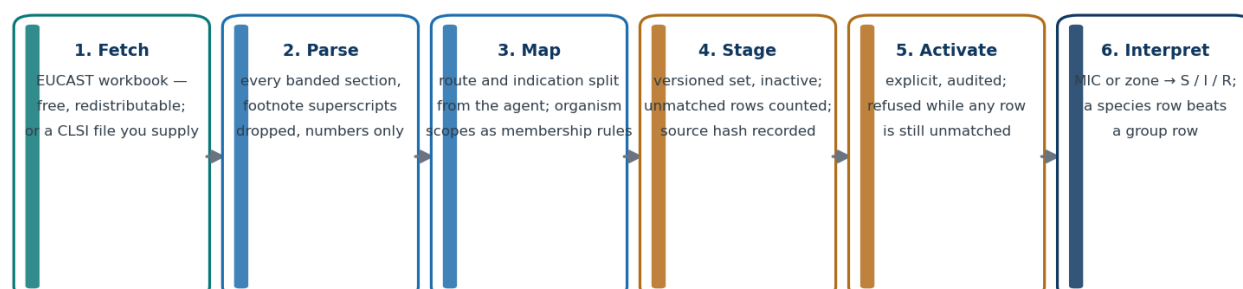
Records are stored in the WHONET code space (organism, antimicrobial and specimen codes) so data is exchangeable with the GLASS ecosystem without a mapping step. Interoperable output is produced on demand: FHIR **Observation** for results, **MeasureReport** for aggregates, HL7 v2 for hospital systems. Organism concepts carry SNOMED CT codes and antimicrobials carry ATC and WHO AWaRe classification, so an indicator can be defined in terms a programme already uses.

Why not a bespoke schema? Because the value of an AMR record is comparability. A national system whose codes are its own becomes an island, and the mapping cost is paid later, by somebody with less context.

23. Breakpoint interpretation

Figure 5. From a published table to an interpreted result

Every stage is refusable, and nothing interprets a result until a person activates it.



Design choices worth stating in a paper

Guidelines are versioned data, not code: a set carries its source hash, edition and per-row provenance, so a result interpreted today can be re-read years later against the table that actually produced it.

Group scopes ("Enterobacterales", "Enterobacterales except Morganellaceae") are predicates over the organism catalogue's taxonomy, not expanded species lists — a laboratory that adds an organism inherits the right breakpoints.

Interpretation refuses rather than guesses: conflicting equally specific rows, or a route the record does not state, leave the result blank with a stated reason instead of picking one.

Figure 5 — from a published table to an interpreted result.

Its properties:

- **Guidelines are versioned data.** A staged set carries its source hash, edition, row provenance and import report, so a result interpreted in 2026 can be re-read in 2030 against the table that actually produced it.

- **Organism scopes are membership rules**, not expanded species lists. EUCAST writes "Enterobacterales", "Coagulase-negative staphylococci", "Enterobacterales except Morganellaceae"; those are stored as predicates over the organism catalogue's taxonomy. A species-specific row always beats a group row for the same organism.
- **Refusal over guessing**. Conflicting equally specific rows, a route the record does not state, or a missing matching row all produce a blank result with a stated reason.
- **Activation is a human act**. A set with unmatched rows cannot be activated at all.

Alternatives considered. Compiling breakpoints into the application makes every guideline revision a software release, which no laboratory can wait for. Accepting a spreadsheet without provenance makes a result unauditible. The staging-plus-activation design is the middle path: data-driven, but never live until somebody takes responsibility for it.

24. Deduplication

Two modes, chosen per analysis and always shown with the result:

- **firstPatientOrganism** — first isolate per patient per organism within the window. This is the CLSI M39 / WHO GLASS convention.
- **allIsolates** — every isolate, for laboratory workload and quality questions.

Why first-isolate is the default. Repeat isolates from the same patient are not independent observations: a single long ICU admission with daily cultures can contribute dozens of resistant isolates and shift a national rate on its own. Reporting resistance without deduplication systematically overestimates it in exactly the settings where the sickest patients are — which is where policy attention lands.

Why the alternative is kept. A laboratory asking "how much work did we do?" or "how complete is our AST?" needs every isolate. The mode is therefore a parameter of the question, not a global setting.

25. Indicators

The priority indicators follow WHO GLASS pathogen-antimicrobial combinations: MRSA; vancomycin-non-susceptible enterococci; third-generation-cephalosporin-non-susceptible *E. coli* and *K. pneumoniae*; carbapenem-non-susceptible Enterobacterales, *P. aeruginosa* and *Acinetobacter* spp.; fluoroquinolone-resistant *E. coli*; penicillin-non-susceptible *S. pneumoniae*.

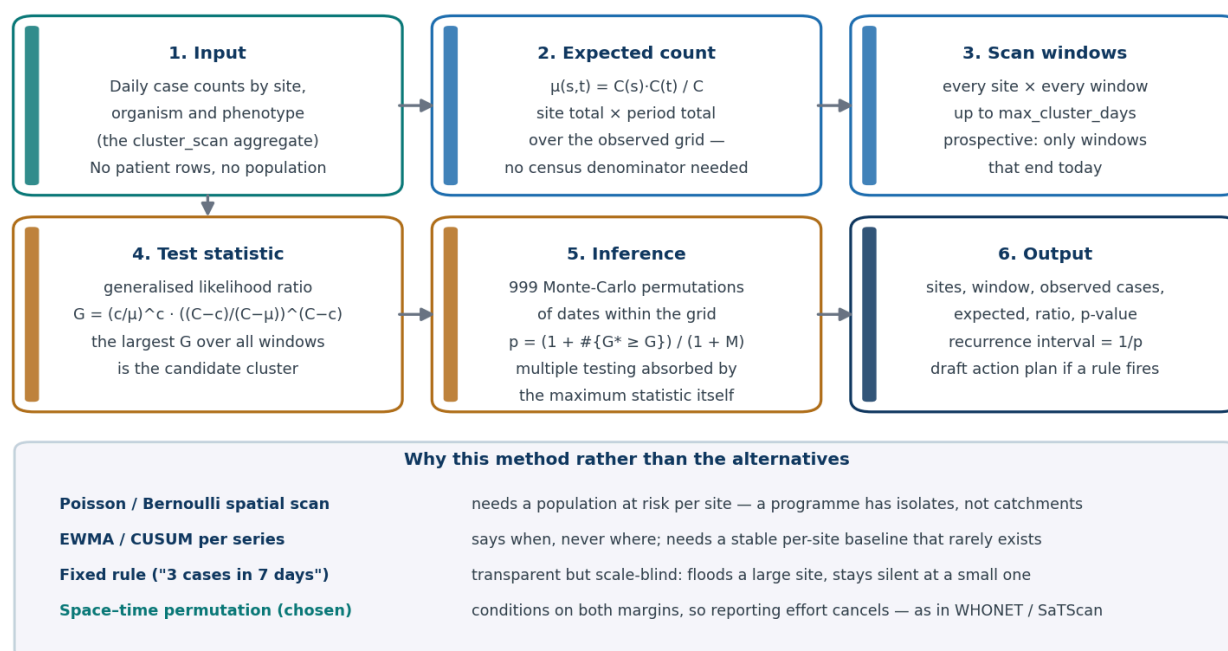
Each is a numerator and a denominator, both reported. Non-susceptible (R + I) is used where GLASS uses it, and the tested-with agent set is stated, because "carbapenem resistance" computed over meropenem alone and over meropenem-or-imipenem-or-ertapenem are different numbers and only one of them matches the neighbouring country's.

26. Outbreak detection

Method: Kulldorff's space-time permutation scan statistic, case-only, prospective by default.

Figure 4. Outbreak signal: space-time permutation scan

Kulldorff's case-only statistic — chosen because an AMR programme has cases, not denominators.

*Figure 4 — the scan pipeline and the alternatives it was chosen over.*

For each candidate cluster — a site (or set of sites) and a time window up to **max_cluster_days** — the expected count is the product of that location's total and that period's total divided by the grand total. The test statistic is the generalised likelihood ratio, and significance comes from Monte-Carlo permutation of dates within the observed grid (999 by default), so the multiple-testing burden of scanning thousands of windows is handled by the distribution of the *maximum* statistic rather than by a correction applied afterwards. Output includes the recurrence interval ($1/p$), which is what an operational programme can act on: "a cluster this extreme would be seen by chance about once every N days".

Why case-only. The Poisson and Bernoulli scan statistics need a population at risk per location. A national AMR programme does not have one: it has isolates from laboratories with different catchments, different testing rates and different referral patterns. Conditioning on both margins removes exactly that — a site that reports twice as much of everything contributes no signal, which is the property that makes the method usable across heterogeneous reporting.

Why not the alternatives.

Method	Why not
Fixed rule ("3 cases in 7 days")	Transparent but scale-blind: floods large sites, silent at small ones
EWMA / CUSUM per series	Detects <i>when</i> but not <i>where</i> ; needs a stable per-site baseline
Poisson / Bernoulli spatial scan	Needs a denominator population AMRIT does not have
Unsupervised clustering	No inferential statement; a cluster is always found

Limitations, stated. The scan detects excess relative to the rest of the observed data, so a uniform national rise is invisible to it — that is what the trend metrics are for. It is sensitive to `baseline_days` and `max_cluster_days`, so both travel with every signal. And it operates on aggregate counts, so it can point a programme at a place and a period, never at a patient.

27. Privacy engineering

Four mechanisms, layered:

1. **Aggregate-only federation.** The wire contract carries counts and rates. A middleware guard rejects any response payload carrying patient identifiers, so a bug in a laboratory build cannot become a national disclosure.
2. **k-anonymity floor.** Cells below `AMRIT_K_ANONYMITY_FLOOR` (default 5) are suppressed before storage, not before display — a suppressed number is never written down.
3. **Coarsening.** Patient residence is stored to a coarsened postal geography; facility coordinates are site-level and consent-gated. Patient coordinates do not exist.
4. **Retention and erasure.** Row-level data expires on the profile's schedule; a subject's rows can be erased on request, and the erasure is itself auditable.

Why not de-identified record sharing? Because de-identified AMR records are re-identifiable in practice: a rare organism, a date and a district are frequently unique. Keeping rows in the laboratory and shipping only answers removes the question rather than mitigating it.

28. Federation protocol design

The laboratory **pulls** questions (`GET /v1/poll` , long-poll, with a WebSocket nudge for immediacy) and posts answers (`POST /v1/respond`). Both credentials — bearer token and out-of-band site token — are required on every request.

Why pull rather than push. A pushing laboratory must hold a queue, a retry policy, a credential for the server and an outbound firewall exception; when it is offline the programme learns nothing about why. A pulling laboratory needs only outbound HTTPS, answers only questions it has allow-listed, and its absence is visible as a missing answer rather than as silence.

Why not a central warehouse. A warehouse is the design that makes every other privacy control a promise. The moment rows are centralised, the protection is policy; while they stay local, it is architecture.

Why the two-token scheme. One token that both authenticates and travels the same channel as enrolment can be replayed by whoever intercepts enrolment. Splitting the factors across two channels means a compromise of either one alone is not enough.

29. Dashboard computation

Metrics are declared in a catalogue (`metrics/catalog.py`) as key, query type, filters and presentation. A refresh dispatches the de-duplicated set of wire queries the catalogue implies, waits for the batch, then computes each metric from *that batch only* and writes a `KPISnapshot` per metric and scope. Snapshots carry their computation time, source and site count, so a dashboard tile can always answer "as of when, from how many sites".

Scopes are canonical (`country` , `admin:<level>` , `site`) and derived from the country profile's tier chain, so a one-tier and a five-tier country use the same code path.

30. Limitations and threats to validity

Stated plainly, because a paper needs them:

- **Denominator heterogeneity.** Resistance rates depend on which isolates a laboratory tests and refers. Coverage metrics (sites reporting, AST completeness) are published alongside the rates for that reason.
 - **Breakpoint version effects.** A guideline revision can move a national rate without any change in biology. Every result carries the set that produced it.
 - **Deduplication sensitivity.** First-isolate versus all-isolate changes rates materially; comparisons across systems must match modes.
 - **Suppression bias.** k-anonymity suppression removes small cells, which are not missing at random — sparse regions lose more.
 - **Signal $\neq$ outbreak.** A scan statistic identifies statistical excess; confirmation is an epidemiological act.
-

Appendix A — Commands

```
# Desktop
pnpm --dir app install && pnpm --dir app dev      # development
pnpm --dir app run check && pnpm --dir app test    # types, lint, tests

# Portal
python3 manage.py migrate
python3 manage.py bootstrap                        # first administrator
python3 manage.py seed_demo [--country IND] [--strict]
python3 manage.py refresh_snapshots --live
python3 manage.py purge_sites --apply              # clears the registry
python3 manage.py purge_dashboard_data --apply     # clears snapshots and results

# Documentation
python3 tools/build_manual_figures.py
AMRIT_PORTAL_URL=... AMRIT_PORTAL_USER=... AMRIT_PORTAL_PASSWORD=... \
  app/node_modules/.bin/electron tools/capture_portal.mjs
python3 tools/build_manual.py
```

Appendix B — Environment

Variable	Meaning
AMRIT_COUNTRY_PROFILE	Country profile id (IN , TESTLAND , ...). Must be passed to the container
AMRIT_SEED_DEMO	1 seeds the demonstration pack — published passwords, demo only
AMRIT_K_ANONYMITY_FLOOR	Minimum cell size before suppression (default 5)
AMRIT_QUERY_TTL_SECONDS	How long a dispatched query stays answerable
AMRIT_LONGPOLL_MAX_WAIT , AMRIT_LONGPOLL_TICK	Long-poll hold time and poll granularity
AMRIT_REFRESH_WAIT_SECONDS	How long a live refresh waits for answers
AMRIT_ENROLMENT_PICKUP_TTL_SECONDS	Lifetime of a one-time pickup token (default 24 h)

Appendix C — Roles and dashboards

Role	Dashboard	Scope
Super administrator, programme administrator	Country	National
Policy maker, public health expert	Country	National
Epidemiologist, researcher	Epidemiology	National, with signals
Administrative health officer	Administrative	The viewer's own unit
Hospital administrator	Hospital	One facility
Press, citizen	Public summary	National, suppressed

Appendix D — Metric catalogue (extract)

`res_*` resistance rates per pathogen-agent pair · `phen_cre`, `phen_mdr` phenotypes ·
`burden_isolates`, `burden_organism_mix`, `burden_specimen_mix` · `antibiogram` ·
`cov_sites_reporting`, `cov_sites_online`, `cov_geo`, `cov_ast_completeness`.

Appendix E — Demonstration credentials

Seeded only by `seed_demo`, with passwords published in this repository. Never enable on a deployment holding real data.

User	Role
<code>superadmin</code>	Super administrator
<code>policy_maker</code>	Policy maker
<code>epidemiologist</code>	Epidemiologist
<code>researcher</code>	Researcher
<code>public_health</code>	Public health expert
<code>state_officer</code> , <code>district_officer</code>	Administrative health officer
<code>hospital_admin</code>	Hospital administrator
<code>press</code> , <code>citizen</code>	Public views